

Roda de Conversa de Dados

Seleção de Features

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Bacharel em Ciência da
Computação (UniFG) - 2014

Mestre em Ciência da
Computação (UFPE) - 2018

Doutorando em Ciência da
computação (UFPE) - 2020

DS & SE @ CESAR - 2017

Professor @ CESAR School -
2022



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Lattes: <http://lattes.cnpq.br/6137784444858483>

Agenda



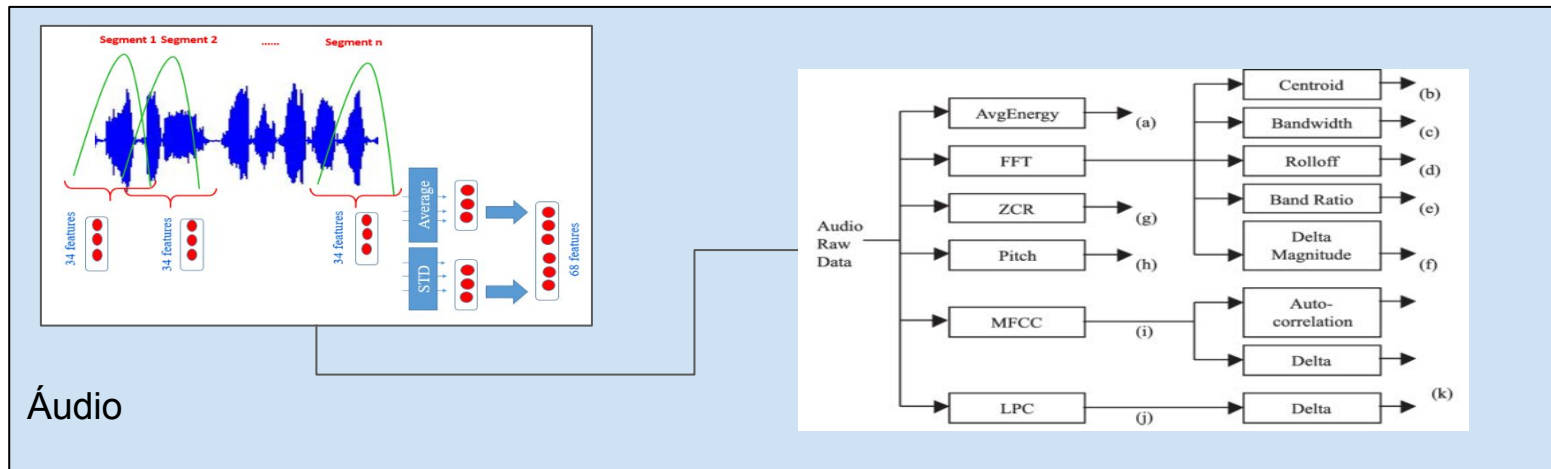
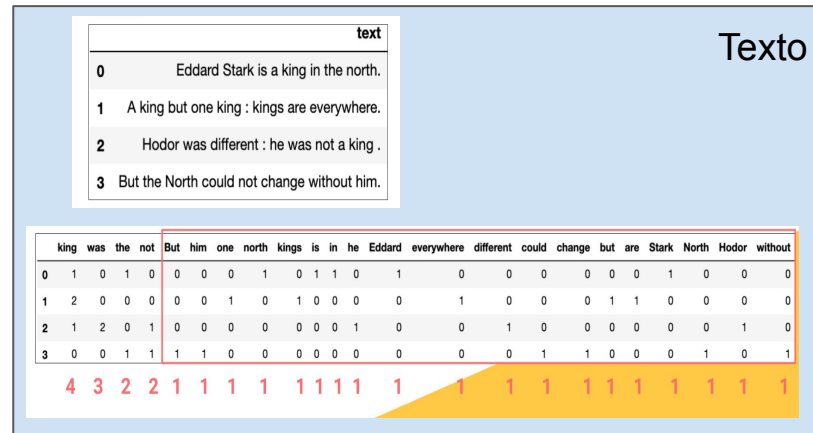
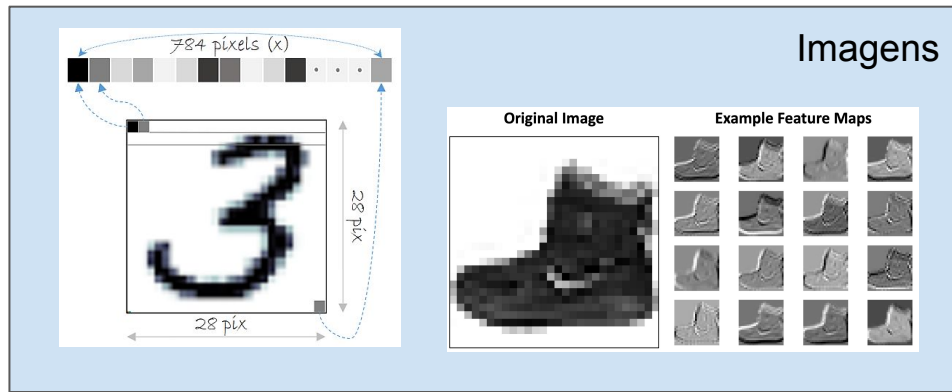
- O que são as features e seus desafios?
- Integração contínua em soluções de data science
- **KNMFS: uma abordagem não supervisionada de seleção de features**
- Experimentos, resultados e trabalhos futuros

O que são features?

- Geralmente os algoritmos de aprendizagem de máquina lidam com uma estrutura matricial de dados (essa matriz é chamada de **dataset**)
- As diversas fontes de dados (db, imagem, áudio, texto, etc) são convertidas para uma única tabela (dataset)
- No **dataset** denominamos as linhas como **exemplos**, **instâncias**, **observações** ou **objetos** e nas colunas como **atributos**, **variáveis** ou **features**

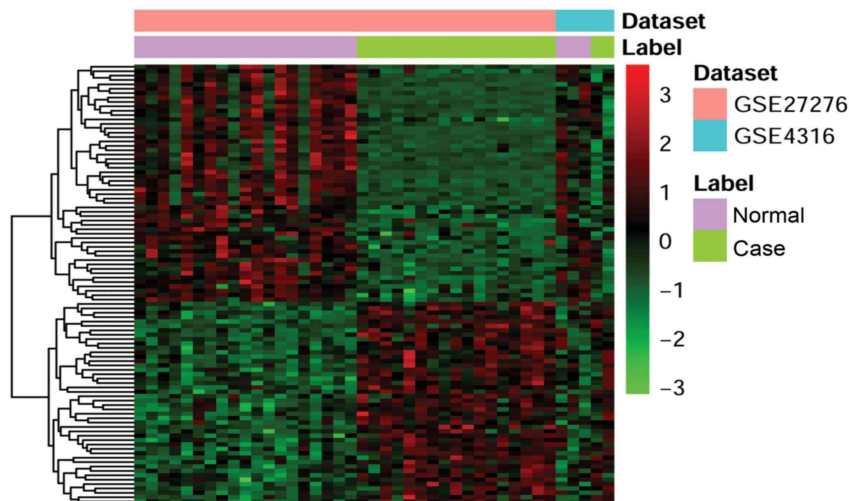
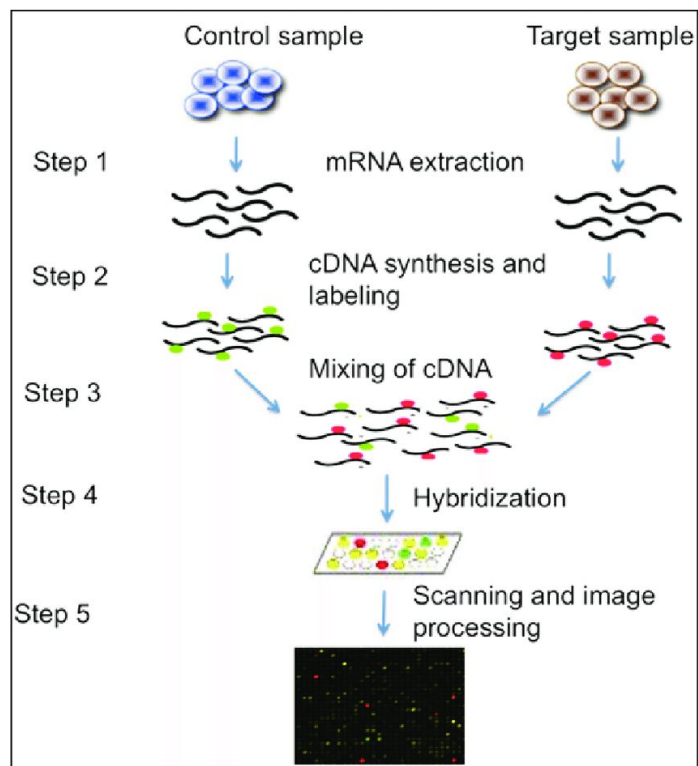
← Features →					Label
Position	Experience	Skill	Country	City	Salary (\$)
Developer	0	1	USA	New York	103100
Developer	1	1	USA	New York	104900
Developer	2	1	USA	New York	106800
Developer	3	1	USA	New York	108700
Developer	4	1	USA	New York	110400
Developer	5	1	USA	New York	112300
Developer	6	1	USA	New York	114200
Developer	7	1	USA	New York	116100
Developer	8	1	USA	New York	117800
Developer	9	1	USA	New York	119700
Developer	10	1	USA	New York	121600

Representação de dados multimídia



Representação dos Dados - Microarray

Através da análise de microarray (dados genéticos) buscamos identificar os genes com maior expressão para uma determinada doença/tumor/estudo.



FENG, Jie; XU, Jing. Identification of pathogenic genes and transcription factors in glaucoma. **Molecular medicine reports**, v. 20, n. 1, p. 216-224, 2019.

Features

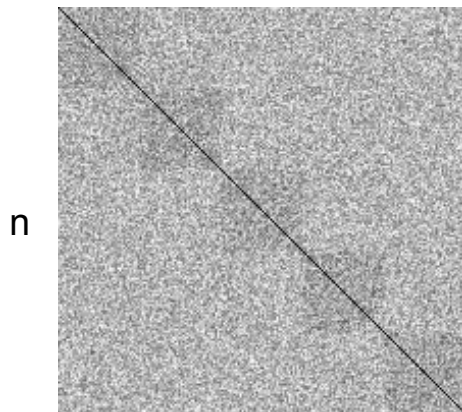
Alguns desafios

Dispersão dos dados

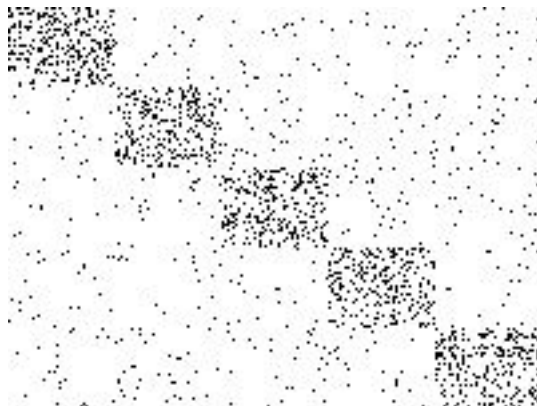
Como diminuir a dispersão dos dados a partir das features?



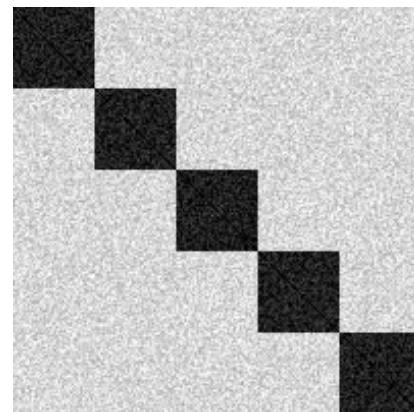
	i	d	j	b	a	e	c	q
id1	0	1	0	1	1	0	1	0
id2	0	1	1	0	1	1	0	0
id3	1	1	0	1	0	0	0	1



n



n = objetos

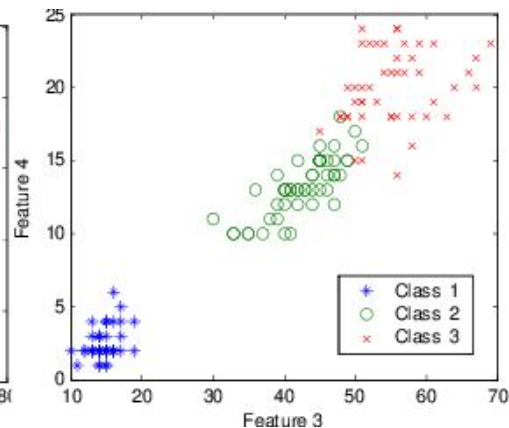
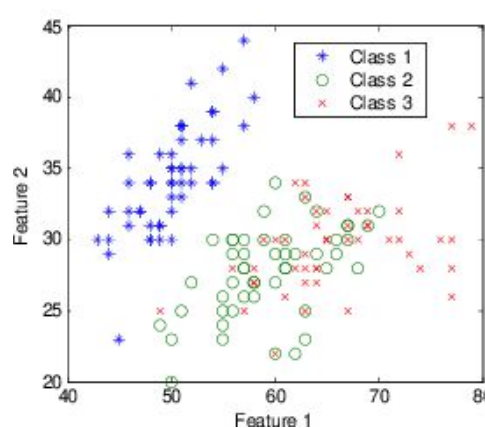
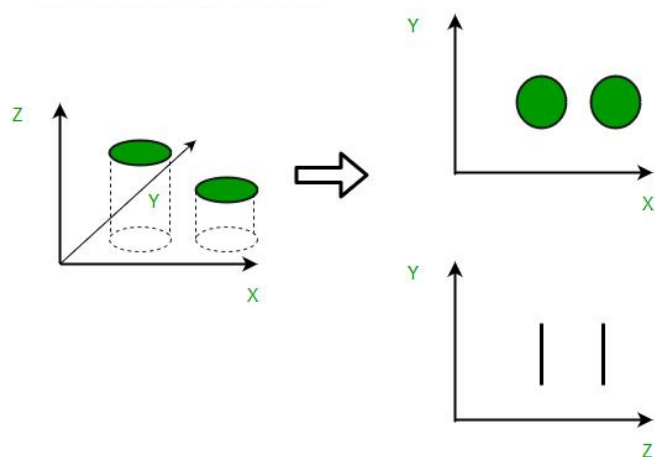


Features

Desafios

Eliminar features indesejadas

Como seleccionar as 2 melhores features ?

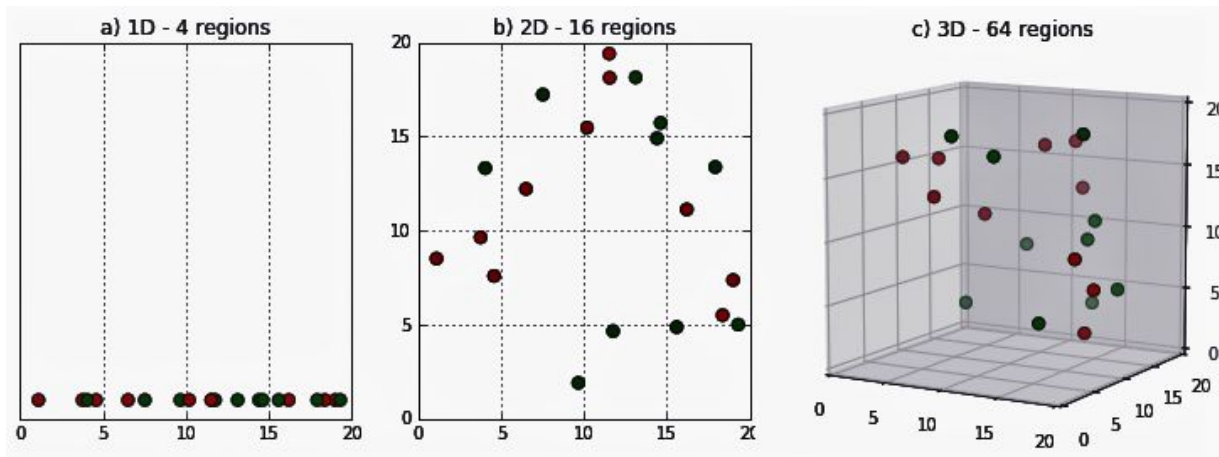


Features

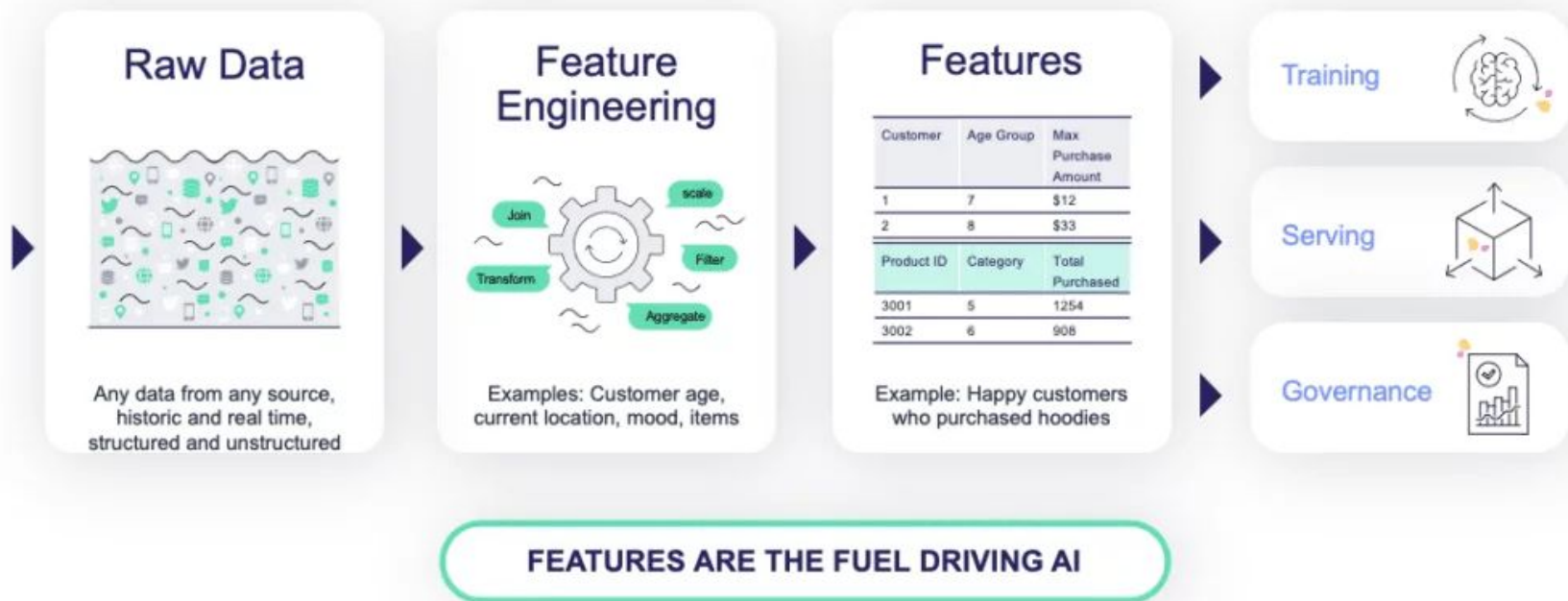
Alguns desafios

Curse of dimensionality

dimensions $\gg$ instances



O que são features?

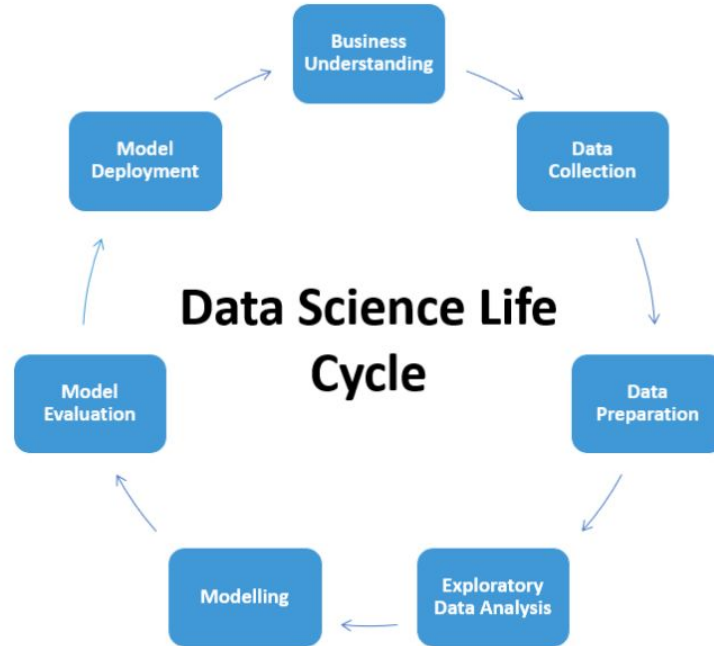


Agenda



- O que são as features e seus desafios?
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Data Science Solution Workflow



<https://laptrinhx.com/complete-life-cycle-of-a-data-science-machine-learning-project-3256392522/>

Continuous Delivery for Machine Learning

Continuous Delivery for Machine Learning (CD4ML) is a **software engineering approach** in which a **cross-functional team** produces machine learning applications based on **code, data, and models** in small and safe increments that can be reproduced and reliably released at any time, in short adaptation cycles.



Danilo Sato

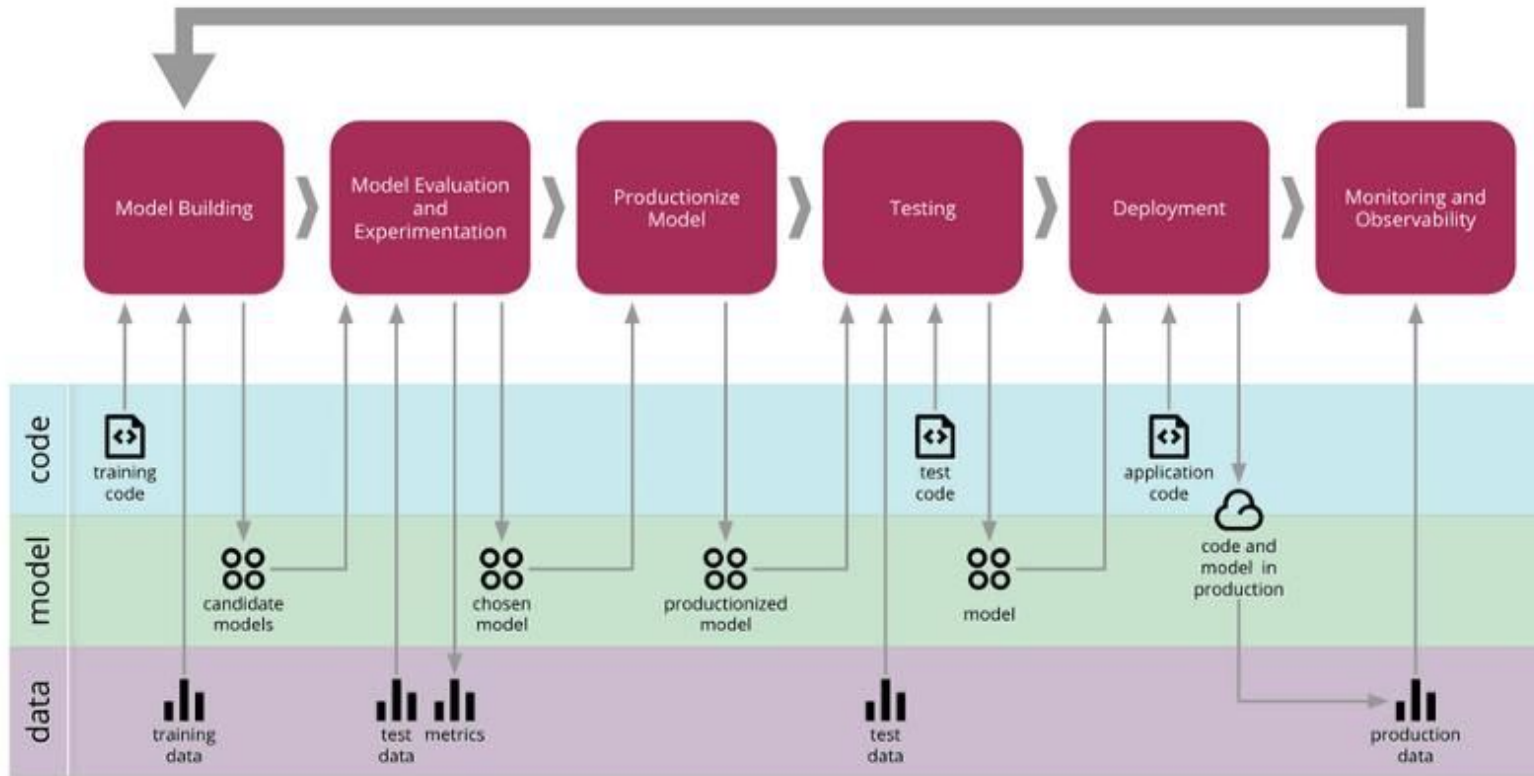


Arif Wider



Christoph Windheuser

CD4ML Process

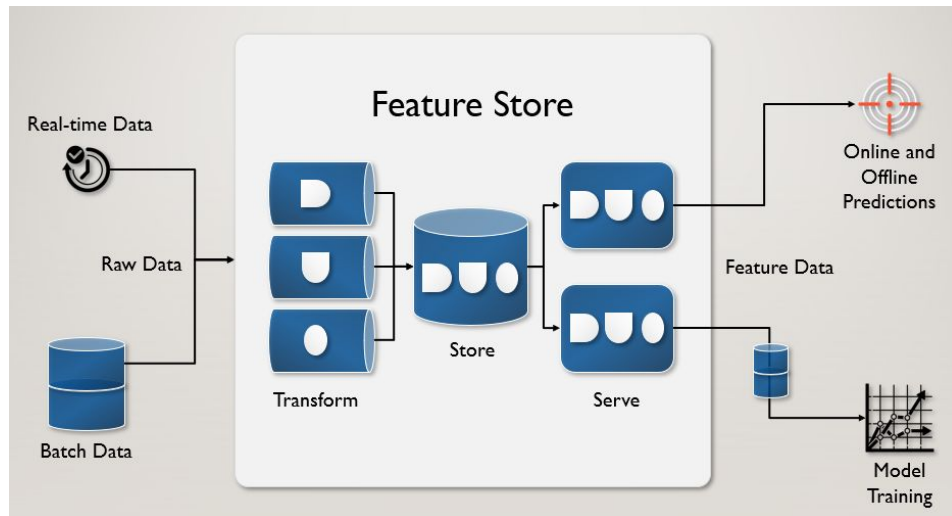


Model Building

Model Building

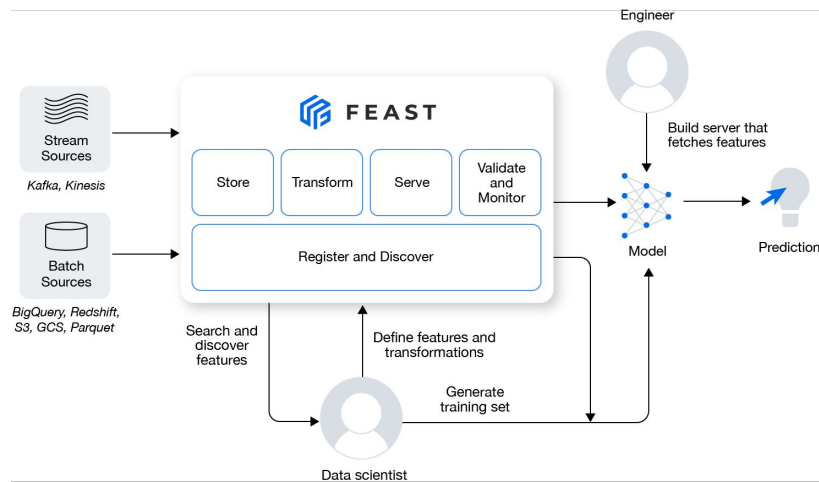


Featurization (Feature Store)



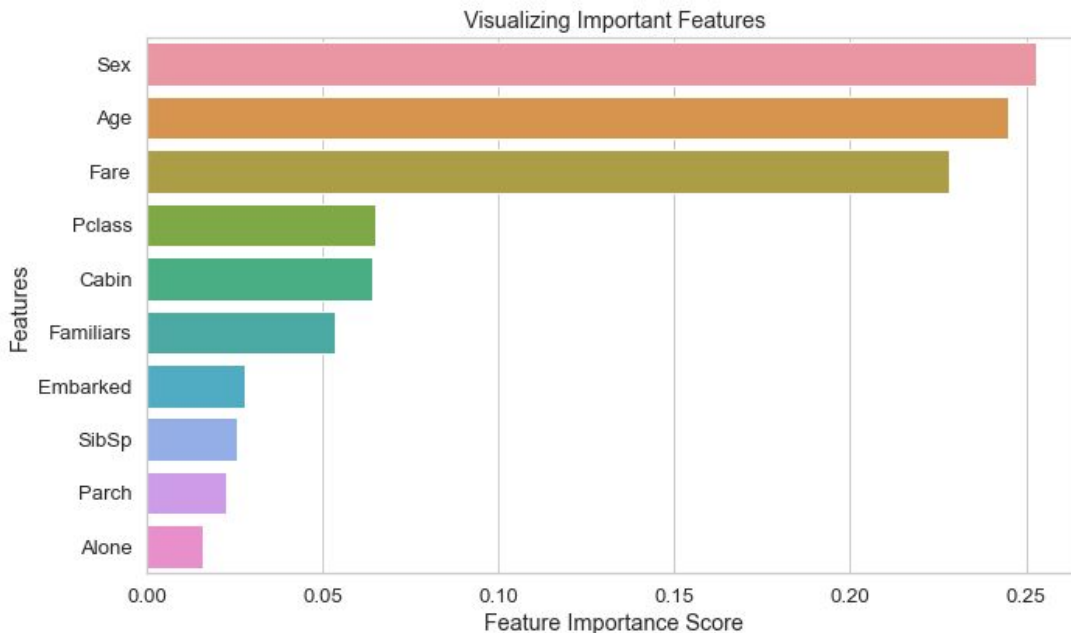
conceito

prática



<https://feast.dev/>

Por que é importante obter features interpretáveis?



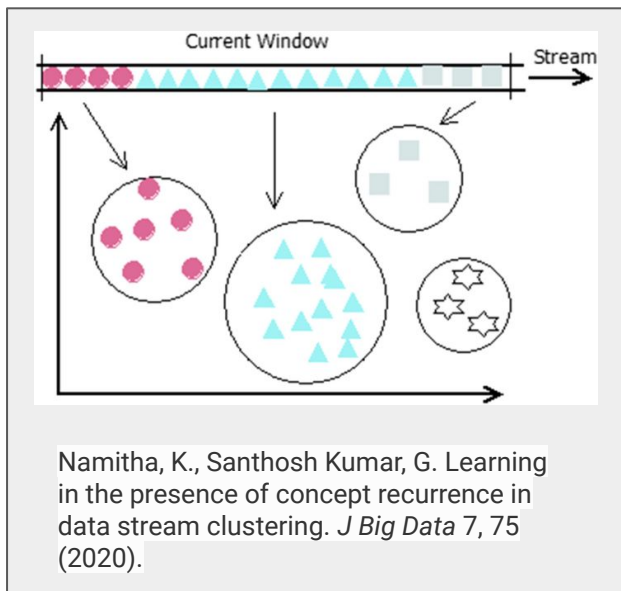
RandomForest evaluation - Titanic



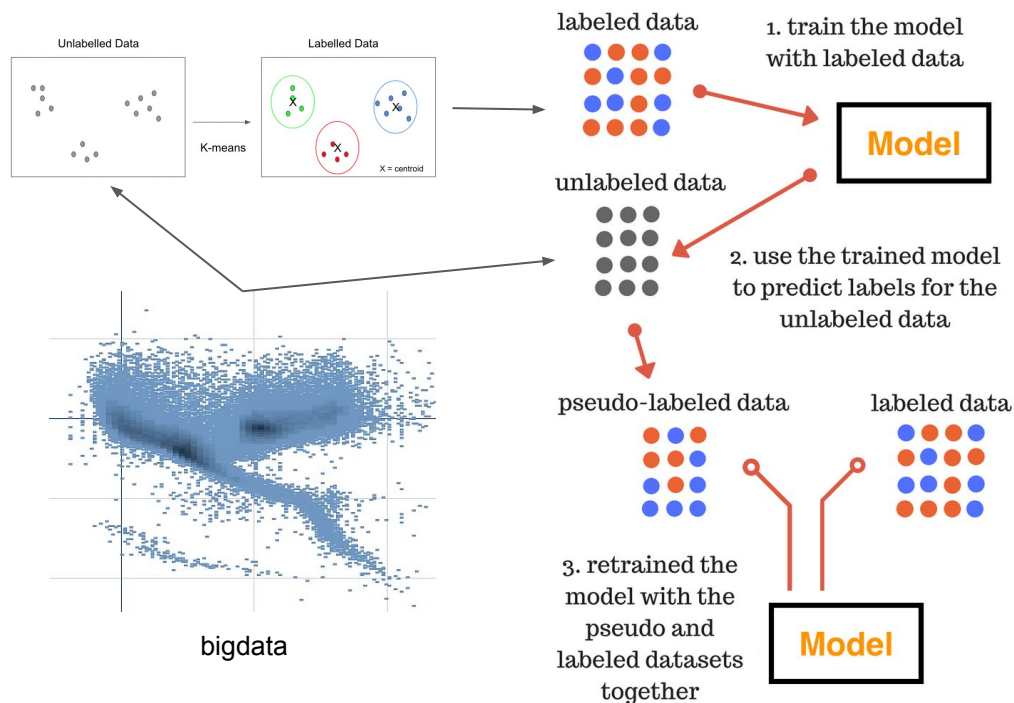
Porque o aprendizado não-supervisionado ?

Unsupervised Learning

Concept drift



Data Labelling



Clustering Solution Workflow

Representação dos
Objetos

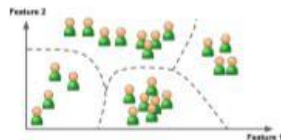
Clusters/
Agrupamentos

**Escolha/Definição
dos Objetos e seus
Atributos**

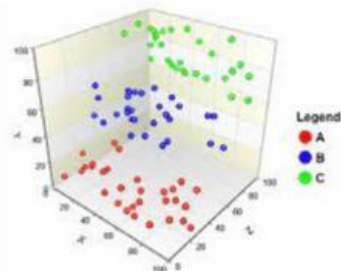


**Exemplos de
features:**
idade x escolaridade
latitude x longitude
altura x peso x idade

**Similaridade entre
Objetos**

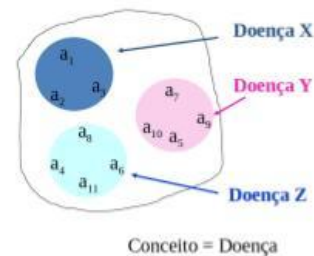


**Algoritmo de
Clustering**



Validação

Interpretação



OLIVEIRA, Marcos de S.; QUEIROZ, Sergio R. de M.; DE CARVALHO, Francisco de AT. **Unsupervised feature selection method based on iterative similarity graph factorization and clustering by modularity**. Expert Systems with Applications, v. 208, p. 118092, 2022.



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Unsupervised feature selection method based on iterative similarity graph factorization and clustering by modularity

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ARTICLE INFO

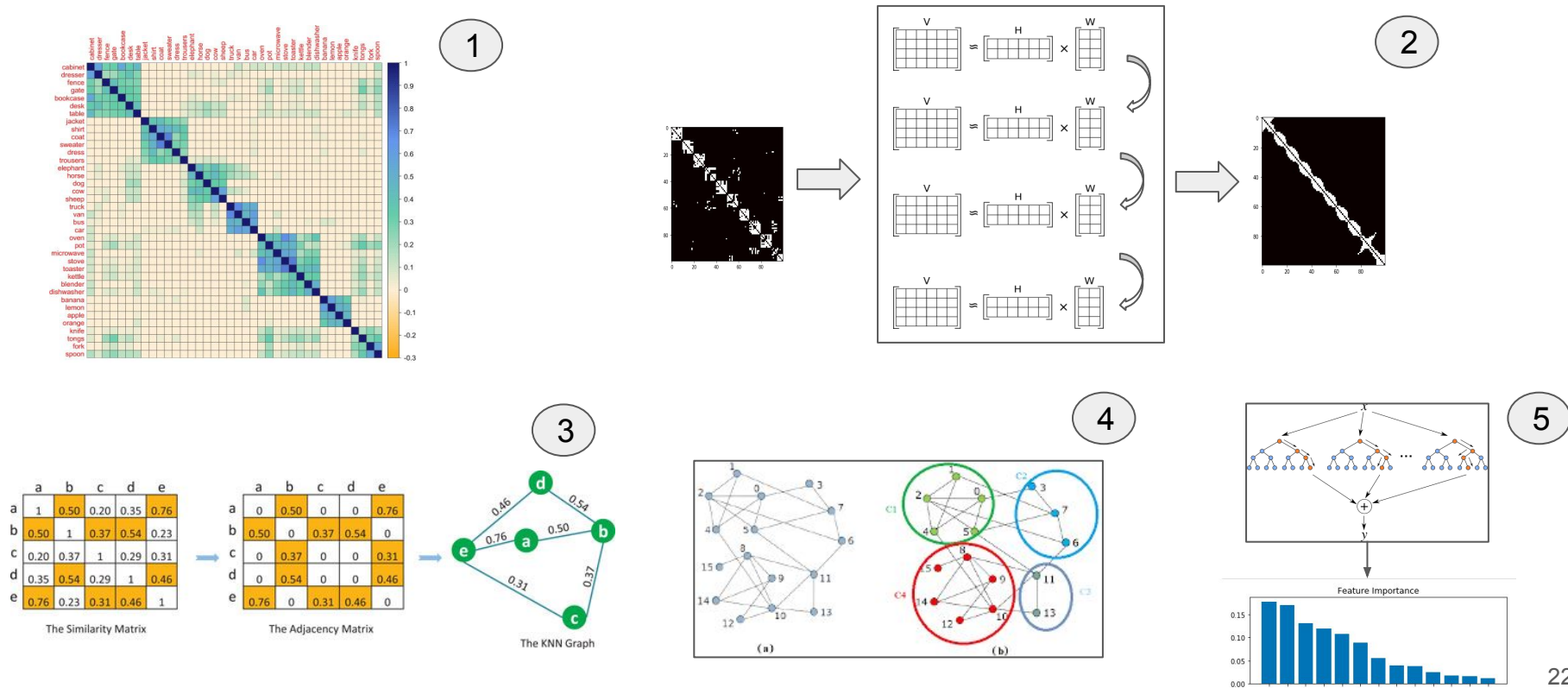
Keywords:

Feature selection
Unsupervised learning
Low-rank approximation
Graph modularity

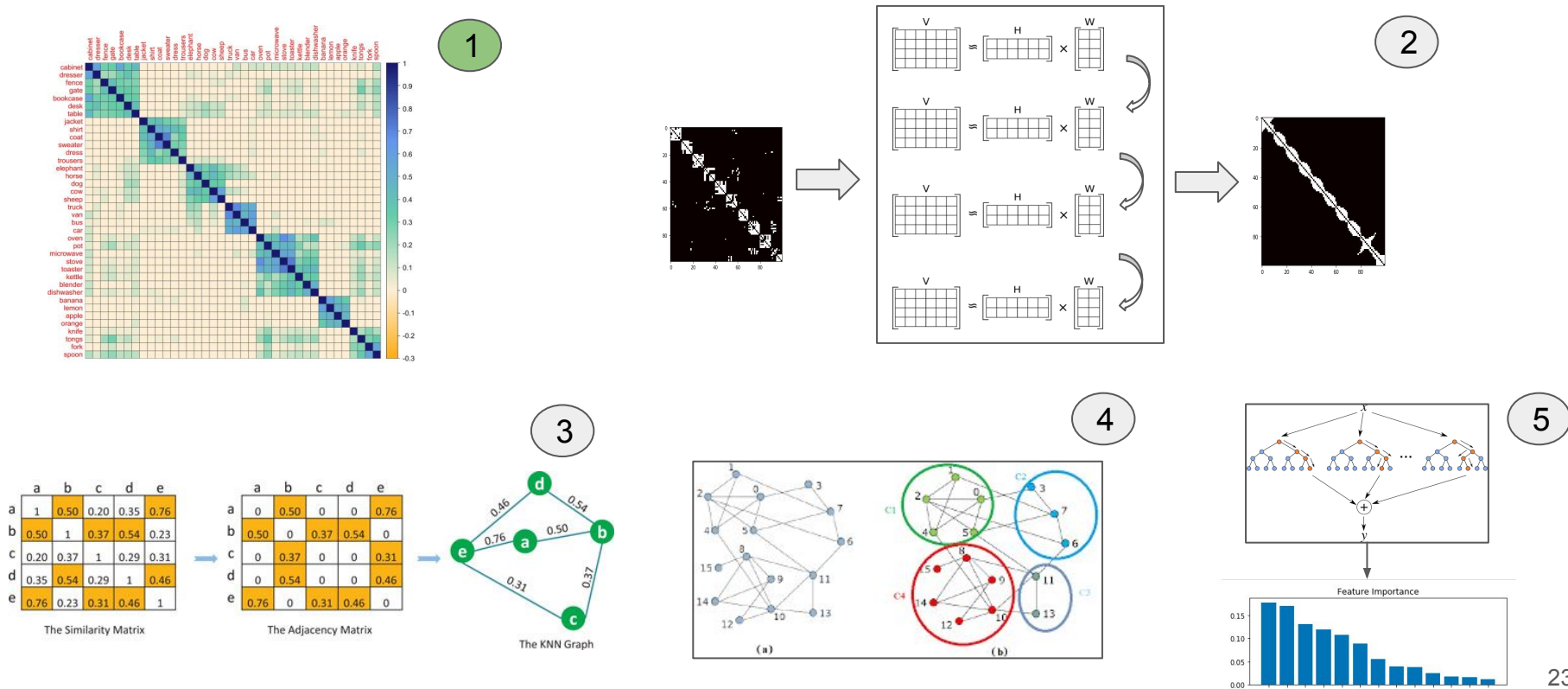
ABSTRACT

Feature selection is an important research area aimed at eliminating unwanted features from high-dimensional datasets. Feature selection methods are categorized according to the availability of labels. Supervised methods usually select features that have high correlations with the data labels, however, this cannot be done by unsupervised methods, since the labels are not available, making the development of them even more challenging. Some unsupervised methods proposed in the literature get around this problem by generating "pseudo-labels", using clustering techniques, and then making feature selection as in the supervised scenario. There are also methods in which the selection is not guided by a clustering criterion, generally performing some kind of reconstruction of the original dataset in low dimensionality. However, in both approaches a local structure is generated, usually starting from a similarity graph built by using the entire set of features. This may compromise the results of the methods when there are many irrelevant and noisy features, which makes it difficult to reveal patterns or an initial representation of the data. Another drawback of some current methods is the large amount of hyper-parameters, or the computational time required for a single execution, which can render such methods unfeasible for some large datasets. In order to address these problems, it is proposed in this work a new unsupervised feature selection method, called KNMFS, which performs the scoring of the features in low computational time using a three-step procedure: (1) a similarity graph learning step based on non-negative iterative matrix factorization together with a Gaussian filter to mitigate the noise effects caused by irrelevant features; (2) a clustering step from the learned graph using a modularity optimization strategy and (3) a random forest algorithm is applied to compute the scores based on the labels generated by the clustering step. To verify the effectiveness of the proposed method, experiments in real and synthetic datasets were conducted. In both cases, the obtained results showed that the KNMFS method, compared to other state-of-the-art methods, obtained good results according to the metrics of Accuracy, ARI and NMI. Friedman's statistical tests were also performed to give stronger evidence to the reported results.

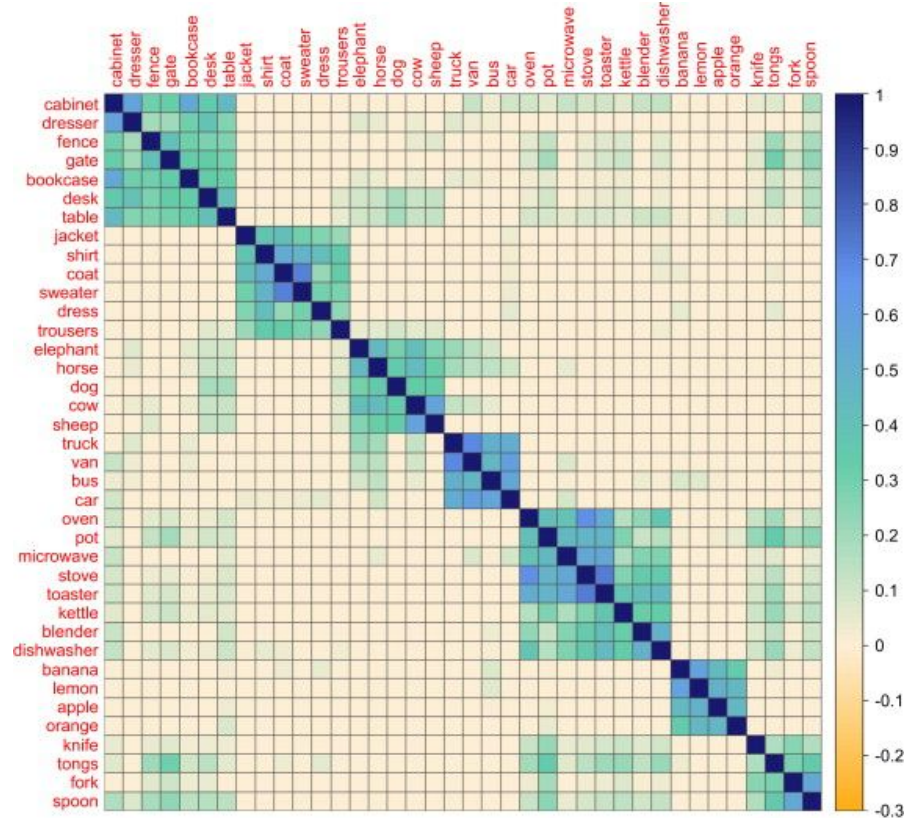
Kernelized NMF and Modularity Feature Selection (KNMFS)



Kernelized NMF and Modularity Feature Selection (KNMFS)



step 1/5: Similarity matrix



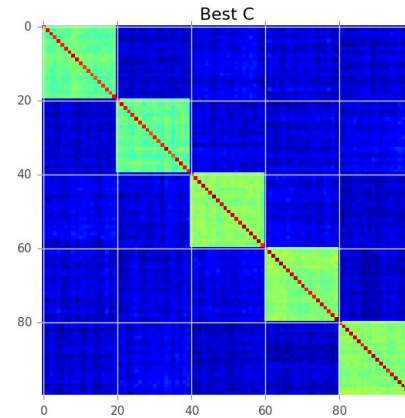
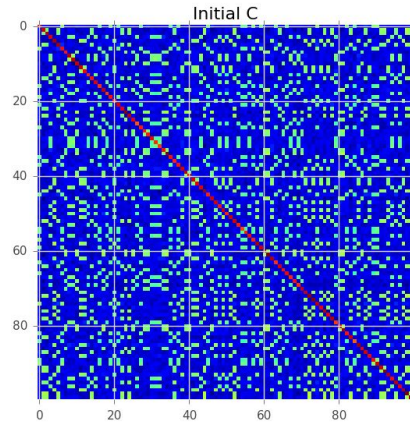
step 1/5: Similarity matrix

0	1	1	0	1
1	0	0	1	0
0	1	1	0	1
1	0	0	1	0
0	1	1	0	1

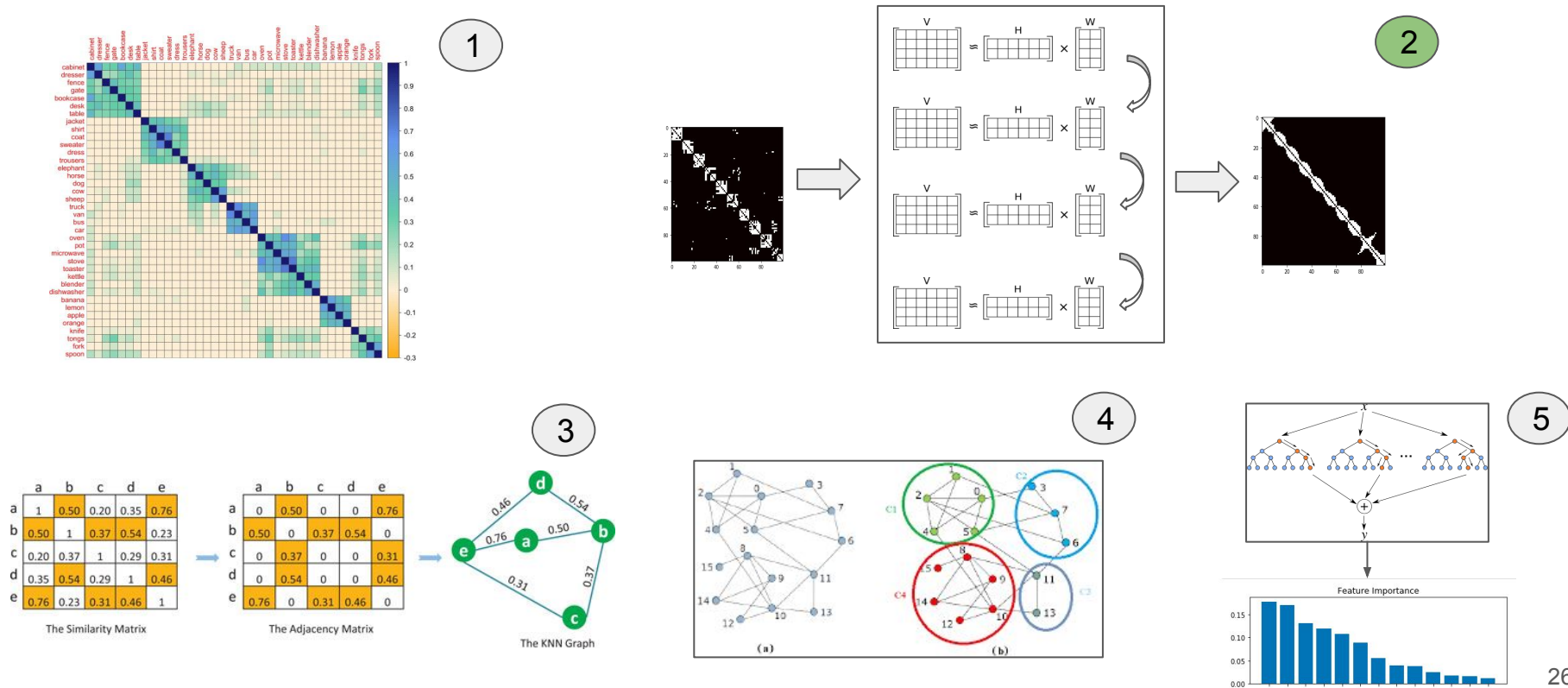
→

1	1	0	0	0
1	1	0	0	0
0	0	1	1	1
0	0	1	1	1
0	0	1	1	1

data sort == grouping



Kernelized NMF and Modularity Feature Selection (KNMFS)

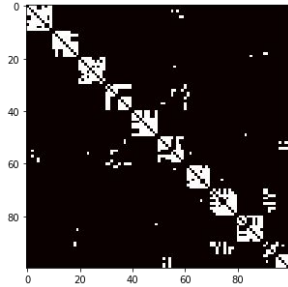


KNMFS

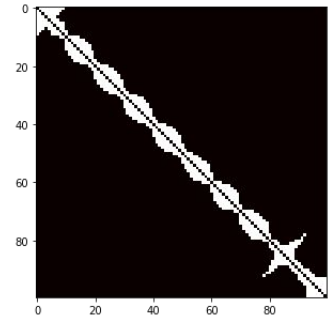
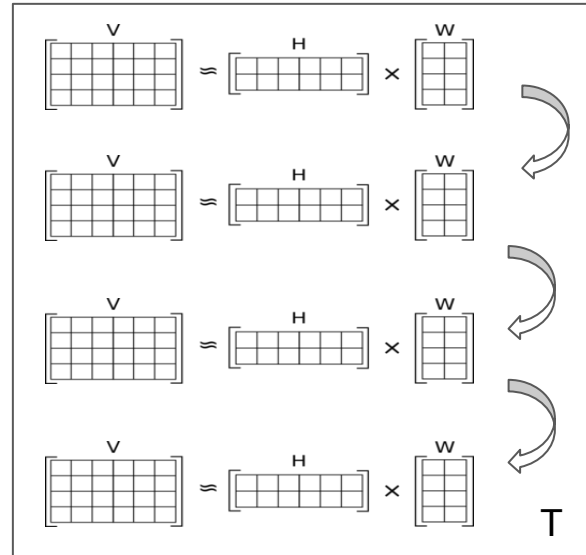
step 2/5: Iterative NMF

$$V \approx H \times W$$

$$G(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{2\sigma^2}}$$

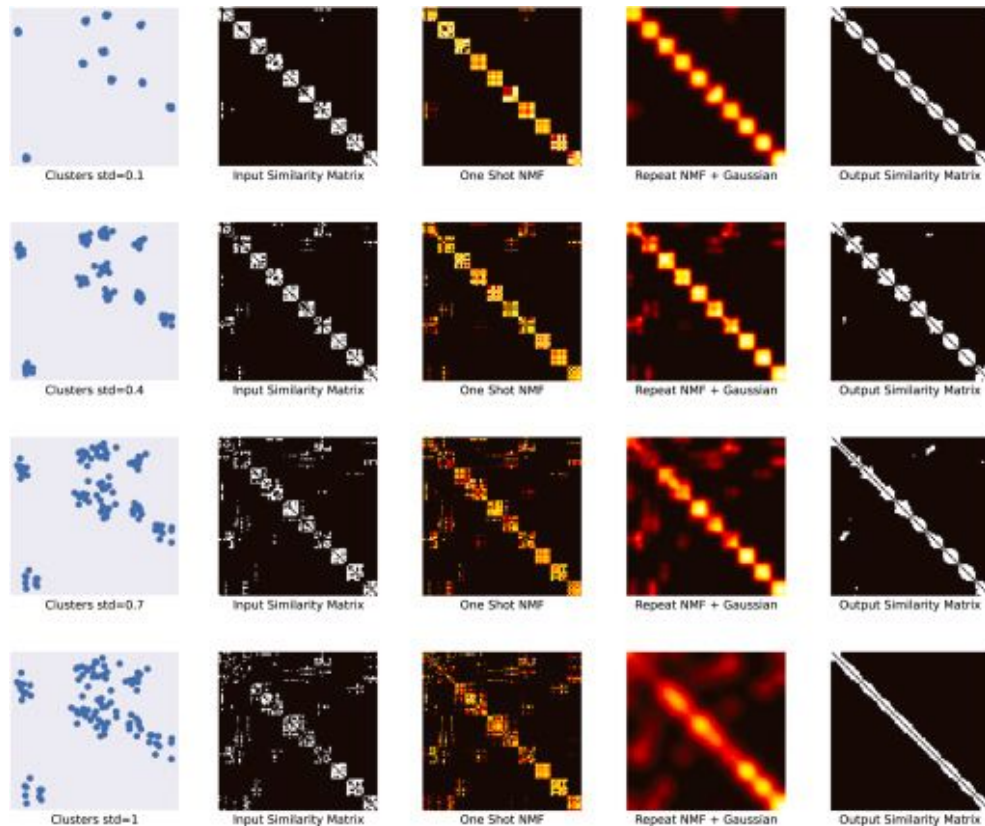


V_initial

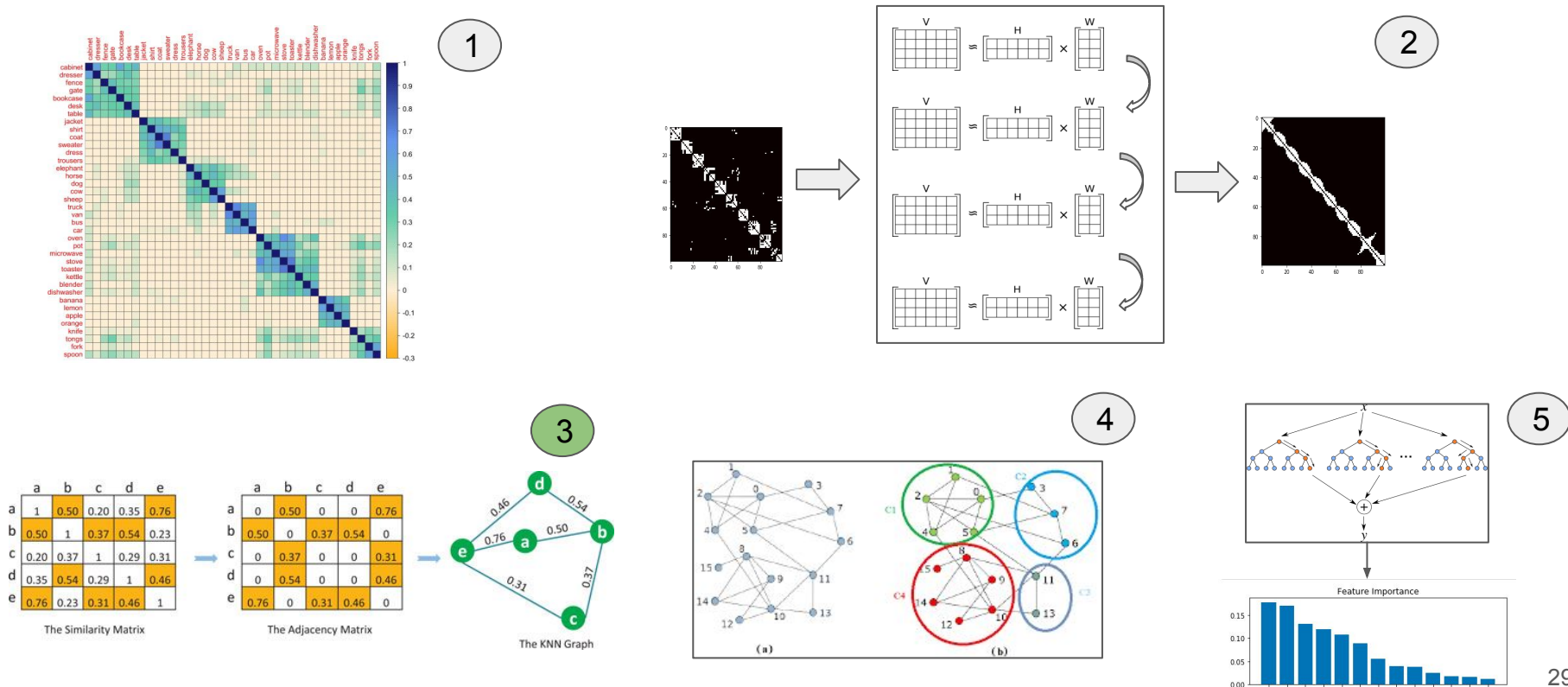


V_final = G(W * H)

step 2/5: Iterative NMF



Kernelized NMF and Modularity Feature Selection (KNMFS)



KNMFS

step 3/5: Similarity Graph

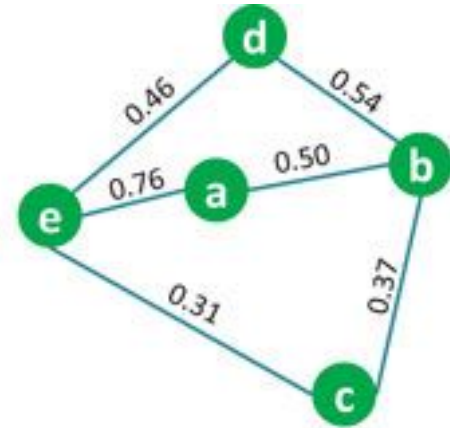
	a	b	c	d	e
a	1	0.50	0.20	0.35	0.76
b	0.50	1	0.37	0.54	0.23
c	0.20	0.37	1	0.29	0.31
d	0.35	0.54	0.29	1	0.46
e	0.76	0.23	0.31	0.46	1

The Similarity Matrix



	a	b	c	d	e
a	0	0.50	0	0	0.76
b	0.50	0	0.37	0.54	0
c	0	0.37	0	0	0.31
d	0	0.54	0	0	0.46
e	0.76	0	0.31	0.46	0

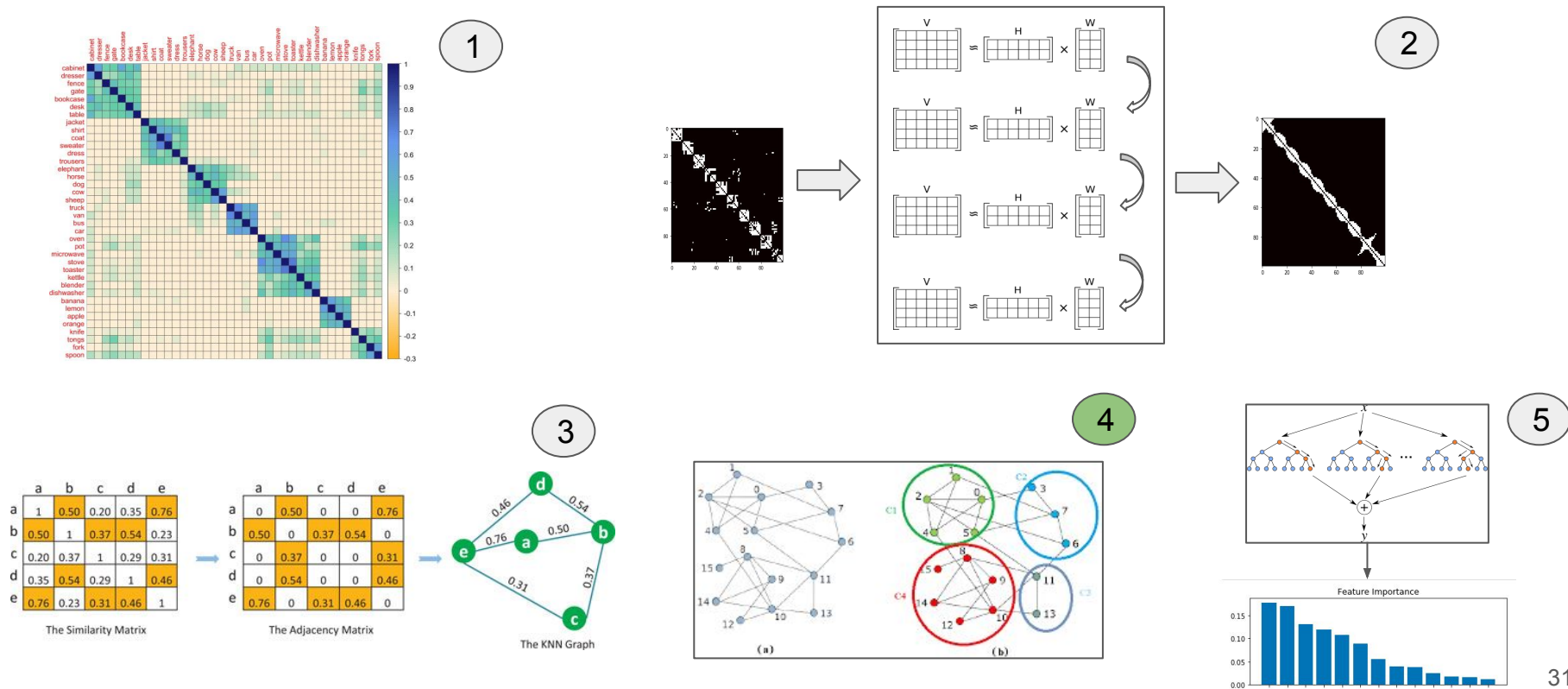
The Adjacency Matrix



The KNN Graph

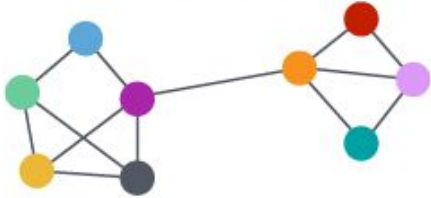
```
from sklearn.neighbors import kneighbors_graph
```

Kernelized NMF and Modularity Feature Selection (KNMFS)

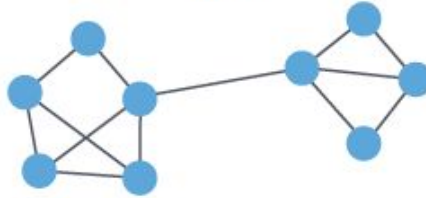


step 4/5: Louvain Community Detection

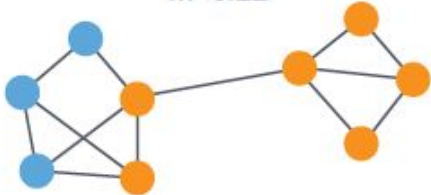
Negative Modularity
 $M=0.12$



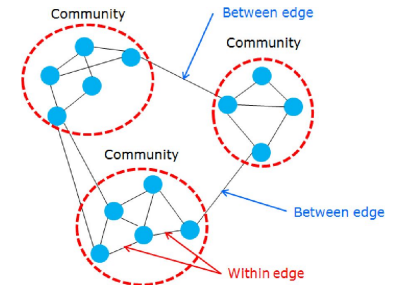
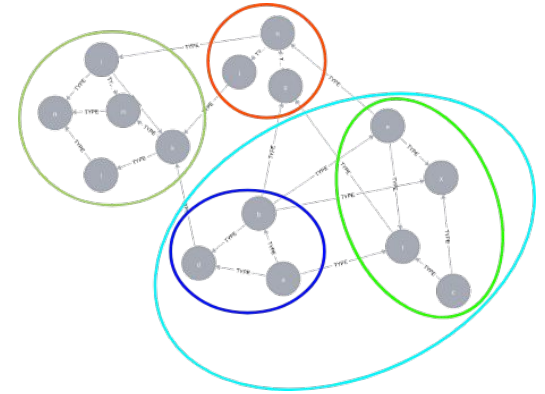
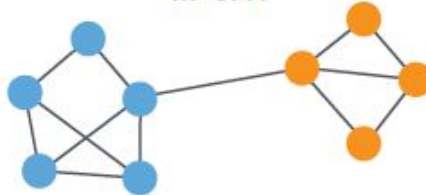
Single Community
 $M=0$



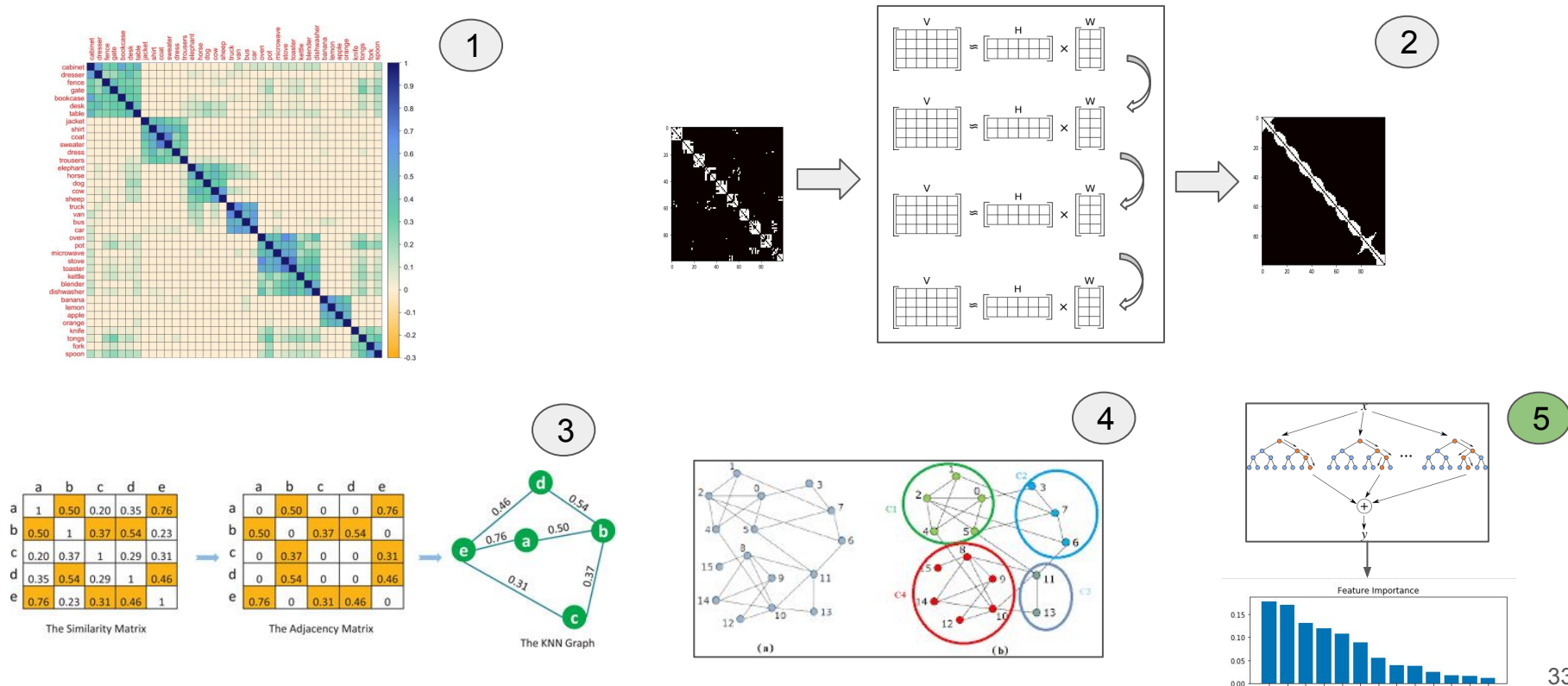
Suboptimal Partition
 $M=0.22$



Optimal Partition
 $M=0.41$



Kernelized NMF and Modularity Feature Selection (KNMFS)



step 5/5: Feature Scoring

	F1	F2	...	Fd
obj1	v1	v2	...	vd
obj2				
objN				

c1	c2	cK
1	0	.
0	1	.
.	.	.



step 5/5: Feature Scoring

	c1	c2	...	cK
F1	s1	s2	...	sK
F2				
Fd				



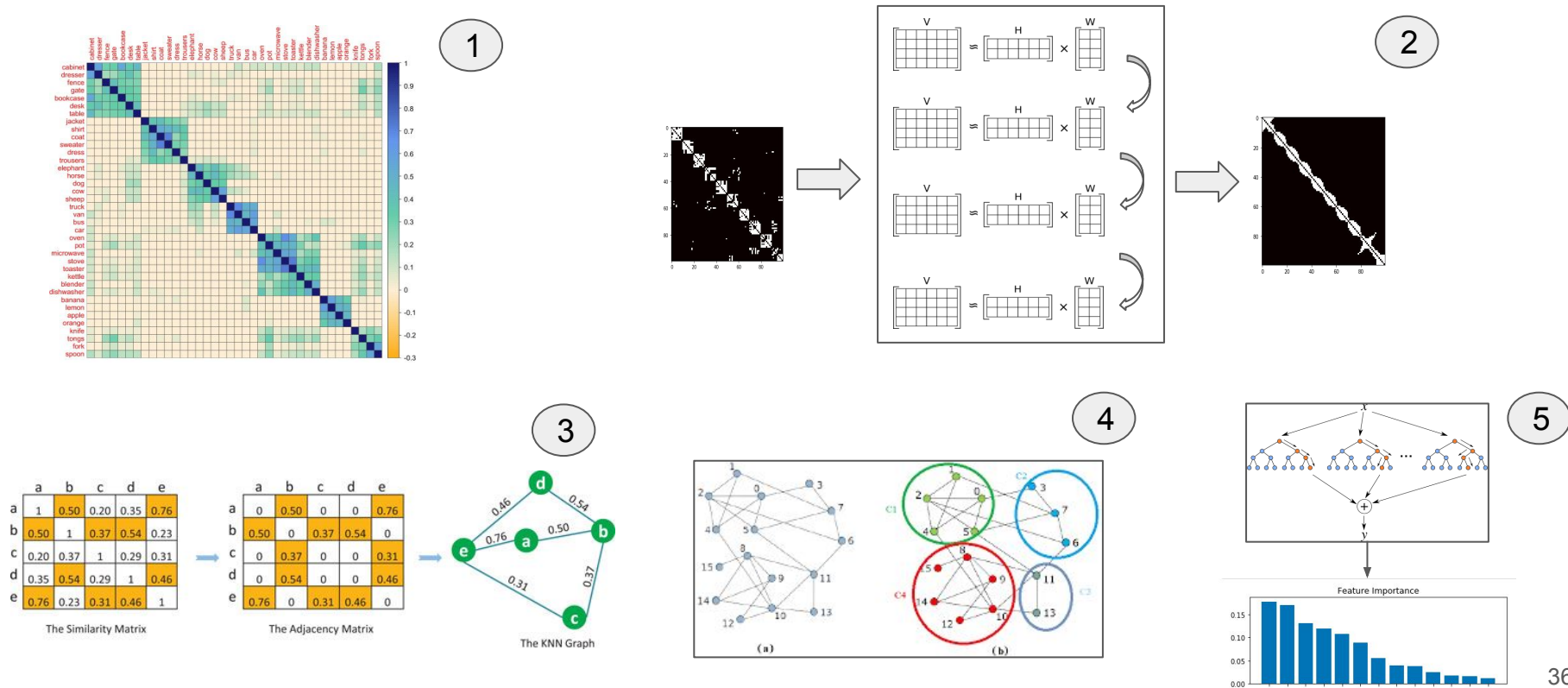
L2_norm



	S
F1	s1
F2	s2
Fd	

$$L_2 = \sqrt{\sum_i^n x_i^2}$$
$$= \sqrt{x_1^2 + x_2^2 + x_3^2 + \dots + x_n^2}$$

Kernelized NMF and Modularity Feature Selection (KNMFS)



KNMFS

<https://github.com/marcosd3souza/KNMFS>

marcosd3souza / KNMFS Public

[Code](#) [Issues](#) [Pull requests](#) [Actions](#) [Projects](#) [Security](#) [Insights](#)

main 1 branch 0 tags

Go to file Code

marcosd3souza paper submission ff7ce4b on Apr 2 11 commits

code	paper submission	7 months ago
LICENSE	Initial commit	13 months ago
README.md	Update README.md	7 months ago

README.md

KNMFS

Unsupervised Feature Selection Method Based on Iterative Similarity Graph Factorization and Clustering by Modularity

Abstract

Feature selection is an important research area aimed at eliminating unwanted features from high-dimensional datasets. Feature selection methods are categorized according to the availability of labels. Supervised methods usually select features that have high correlations with the data labels, however, this cannot be done by unsupervised methods, since the labels are not available, making the development of them even more challenging. Some unsupervised methods proposed in the literature get around this problem by generating "pseudo-labels", using

About

Unsupervised Feature Selection Method Based on Graph Modularity and Data Clustering

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Packages

No packages published

Languages

Python 99.6%

Jupyter Notebook 0.4%





Experimentos

Data & Methods

LS (He et al., 2006)
 SPEC (Zhao & Liu, 2007)
 MCFS (Cai et al., 2010)
 UDFS (Yang et al., 2011)
 JELSR (Gu et al., 2011)
 NDFS (Wang & Zhang, 2012)
 RUFS (Qian & Zhai, 2013)
 GLSPFS (Liu et al., 2013)
 FSASL (Du & Shen, 2015)
 iDetect (Yao et al., 2015)
 SCUFS (Zhu, Zhu et al., 2017)
 GSR_SF (Hu et al., 2017)
 LGRSFS (Zhu, Zhang et al., 2017)
 URAFS (Li et al., 2018)
 DGUFS (Guo & Zhu, 2018)
 AEFS (Han et al., 2018)
 TSFS (Mirzaei et al., 2020)
 SCFS (Parsa et al., 2020)

Table 2
Datasets.

Dataset	Domain	Features	Instances	Classes
warpAR10P	Image	2400	130	10
warpPIE10P	Image	2420	210	10
pixraw10P	Image	10 000	100	10
orlraws10P	Image	10 304	100	10
nci9	Biological	9712	60	9
ProstateGE	Biological	5966	102	2
Carcinom	Biological	9182	174	11
TOX171	Biological	5748	171	4
ALLAML	Biological	7129	72	2
SMK-CAN	Biological	19 993	187	2
isolet	Other	617	1560	26

Metodologia

- Para os métodos foram utilizadas **configurações** default. Por exemplo, $k=5$ para construção do grafo knn e distância euclidiana
- O número de features selecionadas foi através de números mágicos: [10, 20, 30 ... 100]
- O K-Means foi executado com o k igual ao número de classes prévia
- Foram calculadas as métricas de cluster **accuracy**, **NMI** e **ARI**
- Os resultados foram compilados considerando o **MAX**, **AVG** e **STD**
 - em seguida foi realizado o teste de **Friedman** e post-hoc **nemenyi** para identificar uma diferença significativa (CD)

Time

Table 3

Total time spent (in seconds) by the methods of the experiments performed in all Datasets.

Method	warpAR10P	warpPIE10P	pixraw10P	orlraws10P	nci9	ProstateGE	Carcinom	TOX171	ALLAML	SMK-CAN	isolet	Total time
LS	2	3	2	3	2	2	3	2	2	2	6	41
SPEC	1	2	2	2	1	1	2	2	1	2	10	177
MCFS	7	11	57	63	38	22	82	39	25	354	8	1639
UDFS	51	51	2371	7506	9532	2465	3517	1566	2915	128 243	81	179 255
JELSR	171	191	4088	4953	3887	1067	3629	1080	1700	31 695	1190	60 746
NDFS	14	14	550	602	502	126	436	118	209	4568	18	7457
RUFS	556	646	7354	2413	1835	804	2249	1034	974	12 271	1591	43 277
GLSPFS	473	557	501	528	412	273	762	517	282	934	2298	18 545
FSASL	4492	4696	108 368	125 498	102 045	29 935	87 566	26 932	45 911	801 539	11 058	1 463 895
iDetect	75	106	97	103	62	54	200	118	44	455	845	29 949
SCUFS	1090	2890	2600	2625	1077	1654	6671	3920	1039	17 887	62 827	104 280
DGUFs	32	69	86	90	40	50	181	111	36	434	2965	26 037
AEFS	15	17	26	27	20	19	38	25	17	73	30	641
TSFS	86	121	261	275	130	160	388	233	142	981	228	7704
SCFS	240	50	1638	1739	1577	422	1227	349	656	20 088	198	28 184
KNMFS	3	2	4	2	2	2	6	4	2	6	35	68

ACC

Table 4

Clustering results measured by accuracy (Max and Mean ACC% $\pm$ STD) of the compared methods. The best result is highlighted in violet, the second best in blue and the third in cyan. The baseline represents the result when all features are used.

Dataset	Baseline	LS	SPEC	MCFS	UDFS	JELSR	NDFS	RUFS	GLSPFS	FSASL	iDetect	SCUFS	DGUFS	AEFS	TSFS	SCFS	KNMFS
warpAR10P	0,19	0,34 0,56 $\pm$ 0,02	0,34 0,31 $\pm$ 0,02	0,48 0,31 $\pm$ 0,04	0,40 0,42 $\pm$ 0,03	0,27 0,35 $\pm$ 0,01	0,43 0,24 $\pm$ 0,06	0,52 0,32 $\pm$ 0,05	0,41 0,30 $\pm$ 0,03	0,35 0,35 $\pm$ 0,04	0,45 0,26 $\pm$ 0,05	0,39 0,32 $\pm$ 0,05	0,30 0,27 $\pm$ 0,00	0,32 0,29 $\pm$ 0,04	0,24 0,26 $\pm$ 0,02	0,53 0,20 $\pm$ 0,06	0,67 0,31 $\pm$ 0,06
warpPIE10P	0,30	0,44 0,41 $\pm$ 0,02	0,44 0,41 $\pm$ 0,02	0,33 0,29 $\pm$ 0,02	0,64 0,46 $\pm$ 0,13	0,30 0,28 $\pm$ 0,01	0,64 0,49 $\pm$ 0,11	0,57 0,32 $\pm$ 0,04	0,36 0,28 $\pm$ 0,01	0,31 0,22 $\pm$ 0,02	0,50 0,36 $\pm$ 0,00	0,39 0,28 $\pm$ 0,03	0,31 0,24 $\pm$ 0,03	0,28 0,25 $\pm$ 0,00	0,32 0,29 $\pm$ 0,02	0,69 0,38 $\pm$ 0,11	0,46 0,42 $\pm$ 0,02
pixraw10P	0,94	0,93 0,86 $\pm$ 0,10	0,93 0,86 $\pm$ 0,10	0,76 0,69 $\pm$ 0,04	0,85 0,76 $\pm$ 0,04	0,84 0,82 $\pm$ 0,03	0,64 0,51 $\pm$ 0,09	0,97 0,73 $\pm$ 0,10	0,96 0,85 $\pm$ 0,05	0,84 0,80 $\pm$ 0,04	0,77 0,53 $\pm$ 0,01	0,93 0,79 $\pm$ 0,07	0,78 0,67 $\pm$ 0,02	0,92 0,86 $\pm$ 0,08	0,86 0,80 $\pm$ 0,02	0,86 0,63 $\pm$ 0,11	0,98 0,95 $\pm$ 0,04
orlraws10P	0,82	0,73 0,69 $\pm$ 0,03	0,73 0,69 $\pm$ 0,04	0,78 0,64 $\pm$ 0,12	0,77 0,66 $\pm$ 0,05	0,72 0,70 $\pm$ 0,01	0,59 0,45 $\pm$ 0,07	0,87 0,69 $\pm$ 0,07	0,71 0,67 $\pm$ 0,03	0,57 0,46 $\pm$ 0,05	0,81 0,42 $\pm$ 0,01	0,88 0,71 $\pm$ 0,08	0,68 0,60 $\pm$ 0,02	0,77 0,70 $\pm$ 0,02	0,82 0,78 $\pm$ 0,01	0,86 0,58 $\pm$ 0,13	0,84 0,81 $\pm$ 0,02
nci9	0,43	0,38 0,36 $\pm$ 0,01	0,37 0,35 $\pm$ 0,01	0,40 0,36 $\pm$ 0,03	0,43 0,36 $\pm$ 0,05	0,43 0,39 $\pm$ 0,03	0,47 0,32 $\pm$ 0,05	0,53 0,36 $\pm$ 0,05	0,52 0,39 $\pm$ 0,03	0,30 0,24 $\pm$ 0,02	0,42 0,32 $\pm$ 0,00	0,48 0,37 $\pm$ 0,06	0,45 0,40 $\pm$ 0,02	0,48 0,38 $\pm$ 0,02	0,42 0,36 $\pm$ 0,01	0,60 0,37 $\pm$ 0,06	0,62 0,55 $\pm$ 0,04
Prostate-GE	0,03	0,57 0,57 $\pm$ 0,00	0,57 0,57 $\pm$ 0,00	0,58 0,57 $\pm$ 0,01	0,52 0,52 $\pm$ 0,00	0,57 0,54 $\pm$ 0,02	0,88 0,59 $\pm$ 0,11	0,89 0,57 $\pm$ 0,04	0,58 0,55 $\pm$ 0,03	0,53 0,52 $\pm$ 0,00	0,61 0,56 $\pm$ 0,00	0,64 0,56 $\pm$ 0,00	0,61 0,58 $\pm$ 0,01	0,63 0,61 $\pm$ 0,00	0,57 0,57 $\pm$ 0,00	0,62 0,54 $\pm$ 0,03	0,64 0,62 $\pm$ 0,01
Carcinom	0,66	0,60 0,51 $\pm$ 0,03	0,60 0,51 $\pm$ 0,04	0,64 0,54 $\pm$ 0,06	0,57 0,37 $\pm$ 0,11	0,37 0,32 $\pm$ 0,03	0,53 0,38 $\pm$ 0,07	0,73 0,46 $\pm$ 0,09	0,44 0,36 $\pm$ 0,04	0,59 0,41 $\pm$ 0,09	0,64 0,37 $\pm$ 0,01	0,63 0,45 $\pm$ 0,01	0,53 0,47 $\pm$ 0,01	0,60 0,51 $\pm$ 0,02	0,43 0,39 $\pm$ 0,01	0,55 0,39 $\pm$ 0,06	0,81 0,72 $\pm$ 0,06
TOX171	0,25	0,53 0,49 $\pm$ 0,01	0,53 0,50 $\pm$ 0,00	0,44 0,39 $\pm$ 0,04	0,55 0,45 $\pm$ 0,04	0,46 0,44 $\pm$ 0,01	0,46 0,37 $\pm$ 0,03	0,57 0,42 $\pm$ 0,05	0,52 0,44 $\pm$ 0,05	0,46 0,33 $\pm$ 0,04	0,49 0,33 $\pm$ 0,00	0,49 0,43 $\pm$ 0,04	0,49 0,45 $\pm$ 0,01	0,48 0,45 $\pm$ 0,01	0,44 0,42 $\pm$ 0,01	0,50 0,38 $\pm$ 0,04	0,45 0,42 $\pm$ 0,03
ALLAML	0,17	0,75 0,73 $\pm$ 0,00	0,75 0,74 $\pm$ 0,00	0,65 0,55 $\pm$ 0,05	0,74 0,70 $\pm$ 0,02	0,74 0,72 $\pm$ 0,01	0,78 0,59 $\pm$ 0,08	0,90 0,67 $\pm$ 0,08	0,74 0,71 $\pm$ 0,01	0,68 0,62 $\pm$ 0,04	0,74 0,59 $\pm$ 0,00	0,76 0,70 $\pm$ 0,06	0,72 0,67 $\pm$ 0,02	0,75 0,68 $\pm$ 0,03	0,74 0,71 $\pm$ 0,01	0,75 0,63 $\pm$ 0,06	0,79 0,75 $\pm$ 0,03
SMK-CAN	0,00	0,56 0,55 $\pm$ 0,00	0,56 0,55 $\pm$ 0,00	0,59 0,58 $\pm$ 0,01	0,53 0,51 $\pm$ 0,01	0,53 0,53 $\pm$ 0,00	0,61 0,55 $\pm$ 0,03	0,66 0,53 $\pm$ 0,03	0,61 0,54 $\pm$ 0,04	0,65 0,54 $\pm$ 0,03	0,64 0,54 $\pm$ 0,03	0,58 0,53 $\pm$ 0,02	0,53 0,51 $\pm$ 0,01	0,54 0,53 $\pm$ 0,01	0,55 0,52 $\pm$ 0,01	0,62 0,55 $\pm$ 0,03	0,64 0,61 $\pm$ 0,02
isolet	0,76	0,50 0,42 $\pm$ 0,06	0,52 0,42 $\pm$ 0,06	0,48 0,40 $\pm$ 0,06	0,52 0,37 $\pm$ 0,12	0,41 0,31 $\pm$ 0,08	0,38 0,26 $\pm$ 0,06	0,66 0,43 $\pm$ 0,10	0,47 0,31 $\pm$ 0,10	0,64 0,41 $\pm$ 0,09	0,41 0,26 $\pm$ 0,06	0,60 0,39 $\pm$ 0,11	0,10 0,24 $\pm$ 0,41	0,08 0,44 $\pm$ 0,48	0,07 0,39 $\pm$ 0,38	0,66 0,37 $\pm$ 0,14	0,61 0,55 $\pm$ 0,07
AVG Max	0,41	0,58	0,58	0,56	0,59	0,51	0,58	0,72	0,57	0,54	0,59	0,62	0,50	0,53	0,50	0,66	0,68
Mean		0,54	0,54	0,49	0,50	0,48	0,44	0,50	0,50	0,44	0,42	0,50	0,46	0,52	0,49	0,46	0,63

NMI

Table 5

Clustering results measured by Normalized Mutual Information (Max and Mean NMI% $\pm$ STD) of the compared methods. The best result is highlighted in violet, the second best in blue and the third in cyan. The baseline represents the result when all features are used.

Dataset	Baseline	LS	SPEC	MCFS	UDFS	JELSR	NDFS	RUFS	GLSPFS	FSASL	iDetect	SCUFS	DGUFs	AEFS	TSFS	SCFS	KNMFS
warpAR10P	0.19	0.35 0.32 $\pm$ 0.02	0.36 0.32 $\pm$ 0.02	0.46 0.45 $\pm$ 0.01	0.50 0.43 $\pm$ 0.04	0.25 0.21 $\pm$ 0.02	0.48 0.43 $\pm$ 0.04	0.55 0.29 $\pm$ 0.07	0.44 0.36 $\pm$ 0.04	0.42 0.24 $\pm$ 0.07	0.42 0.30 $\pm$ 0.06	0.43 0.24 $\pm$ 0.06	0.30 0.27 $\pm$ 0.02	0.41 0.22 $\pm$ 0.08	0.36 0.29 $\pm$ 0.04	0.63 0.31 $\pm$ 0.09	0.71 0.58 $\pm$ 0.06
warpPIE10P	0.30	0.43 0.41 $\pm$ 0.01	0.43 0.41 $\pm$ 0.01	0.38 0.31 $\pm$ 0.04	0.62 0.56 $\pm$ 0.04	0.30 0.28 $\pm$ 0.01	0.66 0.61 $\pm$ 0.05	0.63 0.33 $\pm$ 0.05	0.34 0.28 $\pm$ 0.02	0.39 0.28 $\pm$ 0.04	0.47 0.37 $\pm$ 0.07	0.36 0.29 $\pm$ 0.03	0.27 0.22 $\pm$ 0.03	0.33 0.29 $\pm$ 0.02	0.38 0.34 $\pm$ 0.02	0.69 0.40 $\pm$ 0.12	0.51 0.47 $\pm$ 0.03
pixraw10P	0.94	0.92 0.89 $\pm$ 0.03	0.92 0.89 $\pm$ 0.03	0.85 0.78 $\pm$ 0.05	0.98 0.88 $\pm$ 0.04	0.92 0.91 $\pm$ 0.01	0.55 0.50 $\pm$ 0.03	0.96 0.82 $\pm$ 0.07	0.95 0.91 $\pm$ 0.02	0.93 0.81 $\pm$ 0.06	0.82 0.59 $\pm$ 0.11	0.96 0.84 $\pm$ 0.06	0.90 0.81 $\pm$ 0.03	0.94 0.89 $\pm$ 0.06	0.94 0.90 $\pm$ 0.03	0.92 0.71 $\pm$ 0.11	0.97 0.96 $\pm$ 0.02
orlraws10P	0.82	0.80 0.79 $\pm$ 0.01	0.80 0.79 $\pm$ 0.01	0.76 0.68 $\pm$ 0.07	0.75 0.71 $\pm$ 0.04	0.85 0.82 $\pm$ 0.02	0.49 0.45 $\pm$ 0.02	0.90 0.75 $\pm$ 0.07	0.85 0.80 $\pm$ 0.04	0.86 0.73 $\pm$ 0.06	0.83 0.44 $\pm$ 0.17	0.90 0.78 $\pm$ 0.06	0.69 0.65 $\pm$ 0.04	0.86 0.79 $\pm$ 0.04	0.91 0.85 $\pm$ 0.06	0.89 0.63 $\pm$ 0.13	0.93 0.89 $\pm$ 0.03
nci9	0.43	0.40 0.37 $\pm$ 0.02	0.40 0.36 $\pm$ 0.02	0.35 0.31 $\pm$ 0.03	0.30 0.24 $\pm$ 0.02	0.49 0.42 $\pm$ 0.04	0.34 0.28 $\pm$ 0.04	0.55 0.36 $\pm$ 0.05	0.48 0.40 $\pm$ 0.03	0.42 0.31 $\pm$ 0.04	0.44 0.29 $\pm$ 0.05	0.50 0.30 $\pm$ 0.08	0.43 0.38 $\pm$ 0.03	0.39 0.31 $\pm$ 0.04	0.39 0.34 $\pm$ 0.03	0.56 0.37 $\pm$ 0.06	0.64 0.57 $\pm$ 0.04
Prostate-GE	0.03	0.01 0.01 $\pm$ 0.00	0.01 0.01 $\pm$ 0.00	0.04 0.03 $\pm$ 0.01	0.08 0.03 $\pm$ 0.02	0.02 0.02 $\pm$ 0.00	0.02 0.02 $\pm$ 0.00	0.39 0.03 $\pm$ 0.03	0.02 0.02 $\pm$ 0.00	0.05 0.02 $\pm$ 0.01	0.07 0.04 $\pm$ 0.01	0.06 0.02 $\pm$ 0.01	0.05 0.04 $\pm$ 0.01	0.05 0.05 $\pm$ 0.01	0.02 0.01 $\pm$ 0.00	0.04 0.02 $\pm$ 0.01	0.07 0.05 $\pm$ 0.02
Carcinom	0.66	0.68 0.58 $\pm$ 0.11	0.65 0.56 $\pm$ 0.12	0.55 0.51 $\pm$ 0.04	0.41 0.30 $\pm$ 0.05	0.42 0.35 $\pm$ 0.03	0.33 0.26 $\pm$ 0.04	0.75 0.46 $\pm$ 0.09	0.47 0.39 $\pm$ 0.04	0.53 0.39 $\pm$ 0.05	0.63 0.31 $\pm$ 0.15	0.73 0.52 $\pm$ 0.07	0.55 0.50 $\pm$ 0.05	0.60 0.53 $\pm$ 0.06	0.43 0.38 $\pm$ 0.03	0.54 0.36 $\pm$ 0.07	0.77 0.70 $\pm$ 0.06
TOX171	0.25	0.38 0.36 $\pm$ 0.01	0.38 0.35 $\pm$ 0.02	0.16 0.08 $\pm$ 0.04	0.39 0.26 $\pm$ 0.06	0.30 0.28 $\pm$ 0.03	0.07 0.05 $\pm$ 0.01	0.38 0.17 $\pm$ 0.07	0.30 0.21 $\pm$ 0.06	0.24 0.13 $\pm$ 0.04	0.23 0.06 $\pm$ 0.07	0.30 0.21 $\pm$ 0.05	0.28 0.23 $\pm$ 0.06	0.27 0.18 $\pm$ 0.06	0.33 0.25 $\pm$ 0.06	0.35 0.10 $\pm$ 0.07	0.37 0.27 $\pm$ 0.08
ALLAML	0.17	0.17 0.15 $\pm$ 0.01	0.17 0.15 $\pm$ 0.01	0.16 0.10 $\pm$ 0.05	0.18 0.13 $\pm$ 0.02	0.15 0.12 $\pm$ 0.01	0.02 0.01 $\pm$ 0.01	0.51 0.10 $\pm$ 0.07	0.15 0.12 $\pm$ 0.02	0.17 0.04 $\pm$ 0.04	0.18 0.05 $\pm$ 0.05	0.23 0.11 $\pm$ 0.05	0.15 0.06 $\pm$ 0.04	0.13 0.07 $\pm$ 0.04	0.13 0.11 $\pm$ 0.02	0.20 0.04 $\pm$ 0.04	0.23 0.17 $\pm$ 0.05
SMK-CAN	0.00	0.01 0.01 $\pm$ 0.00	0.01 0.01 $\pm$ 0.00	0.07 0.03 $\pm$ 0.01	0.01 0.00 $\pm$ 0.00	0.01 0.01 $\pm$ 0.00	0.01 0.00 $\pm$ 0.00	0.10 0.01 $\pm$ 0.01	0.04 0.02 $\pm$ 0.01	0.05 0.02 $\pm$ 0.01	0.06 0.01 $\pm$ 0.01	0.03 0.01 $\pm$ 0.01	0.00 0.00 $\pm$ 0.00	0.01 0.00 $\pm$ 0.00	0.02 0.01 $\pm$ 0.00	0.04 0.01 $\pm$ 0.01	0.06 0.04 $\pm$ 0.02
isolet	0.76	0.68 0.59 $\pm$ 0.06	0.67 0.59 $\pm$ 0.06	0.65 0.52 $\pm$ 0.12	0.71 0.57 $\pm$ 0.13	0.50 0.39 $\pm$ 0.10	0.41 0.29 $\pm$ 0.08	0.74 0.53 $\pm$ 0.10	0.54 0.39 $\pm$ 0.11	0.74 0.56 $\pm$ 0.11	0.58 0.34 $\pm$ 0.09	0.66 0.48 $\pm$ 0.11	0.63 0.53 $\pm$ 0.12	0.70 0.62 $\pm$ 0.09	0.63 0.58 $\pm$ 0.06	0.76 0.46 $\pm$ 0.16	0.75 0.68 $\pm$ 0.08
AVG Max	0.32	0.35	0.35	0.32	0.35	0.30	0.24	0.47	0.33	0.34	0.35	0.47	0.31	0.34	0.33	0.51	0.55
Mean		0.32	0.32	0.27	0.29	0.27	0.21	0.28	0.28	0.25	0.20	0.35	0.27	0.28	0.29	0.31	0.49

ARI

Table 6

Clustering results measured by Adjusted Rand Index (Max and Mean ARI% $\pm$ STD) of the compared methods. The best result is highlighted in violet, the second best in blue and the third in cyan. The baseline represents the result when all features are used.

Dataset	Baseline	LS	SPEC	MCFS	UDFS	JELSR	NDFS	RUFS	GLSPFS	FSASL	iDetect	SCUFS	DGUFS	AEFS	TSFS	SCFS	KNMFS
warpARIOP	0,02	0,12 0,09 $\pm$ 0,02	0,12 0,09 $\pm$ 0,02	0,24 0,21 $\pm$ 0,02	0,24 0,17 $\pm$ 0,03	0,06 0,04 $\pm$ 0,01	0,20 0,17 $\pm$ 0,04	0,31 0,09 $\pm$ 0,05	0,20 0,14 $\pm$ 0,03	0,19 0,06 $\pm$ 0,04	0,19 0,08 $\pm$ 0,05	0,20 0,06 $\pm$ 0,04	0,07 0,06 $\pm$ 0,01	0,19 0,05 $\pm$ 0,05	0,14 0,08 $\pm$ 0,03	0,39 0,10 $\pm$ 0,06	0,50 0,37 $\pm$ 0,07
warpPIE1OP	0,08	0,17 0,16 $\pm$ 0,01	0,17 0,16 $\pm$ 0,01	0,14 0,09 $\pm$ 0,03	0,32 0,26 $\pm$ 0,04	0,11 0,09 $\pm$ 0,01	0,44 0,37 $\pm$ 0,06	0,39 0,12 $\pm$ 0,04	0,16 0,09 $\pm$ 0,01	0,15 0,09 $\pm$ 0,03	0,25 0,14 $\pm$ 0,05	0,18 0,10 $\pm$ 0,02	0,07 0,05 $\pm$ 0,01	0,13 0,09 $\pm$ 0,02	0,14 0,12 $\pm$ 0,01	0,48 0,18 $\pm$ 0,11	0,27 0,23 $\pm$ 0,03
pixraw10P	0,87	0,86 0,79 $\pm$ 0,07	0,86 0,79 $\pm$ 0,07	0,68 0,58 $\pm$ 0,07	0,96 0,72 $\pm$ 0,10	0,80 0,79 $\pm$ 0,02	0,31 0,23 $\pm$ 0,05	0,93 0,64 $\pm$ 0,12	0,91 0,80 $\pm$ 0,05	0,83 0,61 $\pm$ 0,10	0,64 0,35 $\pm$ 0,13	0,93 0,67 $\pm$ 0,11	0,76 0,60 $\pm$ 0,06	0,87 0,77 $\pm$ 0,10	0,84 0,77 $\pm$ 0,05	0,81 0,49 $\pm$ 0,14	0,96 0,92 $\pm$ 0,04
orlraws10P	0,65	0,61 0,56 $\pm$ 0,03	0,61 0,56 $\pm$ 0,03	0,58 0,46 $\pm$ 0,09	0,49 0,41 $\pm$ 0,07	0,67 0,63 $\pm$ 0,03	0,24 0,19 $\pm$ 0,03	0,78 0,55 $\pm$ 0,10	0,67 0,60 $\pm$ 0,06	0,71 0,51 $\pm$ 0,08	0,69 0,20 $\pm$ 0,19	0,81 0,59 $\pm$ 0,10	0,46 0,40 $\pm$ 0,03	0,74 0,63 $\pm$ 0,07	0,80 0,70 $\pm$ 0,10	0,79 0,39 $\pm$ 0,17	0,83 0,76 $\pm$ 0,05
nci9	0,17	0,09 0,07 $\pm$ 0,01	0,09 0,07 $\pm$ 0,01	0,09 0,04 $\pm$ 0,03	0,02 0,00 $\pm$ 0,01	0,18 0,12 $\pm$ 0,03	0,08 0,03 $\pm$ 0,03	0,27 0,08 $\pm$ 0,05	0,18 0,10 $\pm$ 0,03	0,14 0,03 $\pm$ 0,03	0,14 0,03 $\pm$ 0,04	0,22 0,06 $\pm$ 0,05	0,14 0,08 $\pm$ 0,03	0,12 0,04 $\pm$ 0,04	0,10 0,05 $\pm$ 0,03	0,31 0,09 $\pm$ 0,06	0,40 0,31 $\pm$ 0,05
Prostate-GE	0,02	0,01 0,01 $\pm$ 0,00	0,01 0,01 $\pm$ 0,00	0,05 0,03 $\pm$ 0,01	0,03 0,00 $\pm$ 0,01	0,01 0,00 $\pm$ 0,00	0,00 0,00 $\pm$ 0,00	0,47 0,03 $\pm$ 0,04	0,02 0,01 $\pm$ 0,01	0,05 0,02 $\pm$ 0,01	0,04 0,01 $\pm$ 0,01	0,05 0,01 $\pm$ 0,01	0,03 0,02 $\pm$ 0,01	0,05 0,04 $\pm$ 0,01	0,02 0,01 $\pm$ 0,00	0,05 0,01 $\pm$ 0,01	0,07 0,05 $\pm$ 0,01
Carcinom	0,52	0,50 0,39 $\pm$ 0,11	0,50 0,36 $\pm$ 0,13	0,44 0,36 $\pm$ 0,06	0,14 0,06 $\pm$ 0,03	0,20 0,14 $\pm$ 0,03	0,17 0,10 $\pm$ 0,04	0,64 0,29 $\pm$ 0,11	0,24 0,17 $\pm$ 0,04	0,39 0,20 $\pm$ 0,06	0,49 0,15 $\pm$ 0,15	0,66 0,35 $\pm$ 0,09	0,39 0,33 $\pm$ 0,06	0,45 0,38 $\pm$ 0,07	0,22 0,18 $\pm$ 0,03	0,38 0,17 $\pm$ 0,07	0,64 0,54 $\pm$ 0,08
TOX171	0,20	0,22 0,20 $\pm$ 0,01	0,22 0,20 $\pm$ 0,01	0,11 0,05 $\pm$ 0,03	0,30 0,18 $\pm$ 0,05	0,19 0,17 $\pm$ 0,02	0,04 0,02 $\pm$ 0,01	0,27 0,11 $\pm$ 0,06	0,22 0,15 $\pm$ 0,05	0,16 0,08 $\pm$ 0,03	0,16 0,03 $\pm$ 0,05	0,21 0,14 $\pm$ 0,04	0,20 0,16 $\pm$ 0,04	0,17 0,11 $\pm$ 0,04	0,21 0,17 $\pm$ 0,04	0,26 0,06 $\pm$ 0,05	0,20 0,14 $\pm$ 0,05
ALLAML	0,24	0,24 0,21 $\pm$ 0,01	0,24 0,21 $\pm$ 0,02	0,08 0,03 $\pm$ 0,03	0,15 0,10 $\pm$ 0,02	0,21 0,17 $\pm$ 0,02	0,02 −0,01 $\pm$ 0,01	0,64 0,12 $\pm$ 0,10	0,21 0,17 $\pm$ 0,02	0,26 0,06 $\pm$ 0,05	0,21 0,00 $\pm$ 0,05	0,33 0,14 $\pm$ 0,08	0,21 0,10 $\pm$ 0,06	0,18 0,11 $\pm$ 0,05	0,18 0,16 $\pm$ 0,03	0,24 0,04 $\pm$ 0,07	0,33 0,24 $\pm$ 0,06
SMK-CAN	0,00	0,01 0,01 $\pm$ 0,00	0,01 0,01 $\pm$ 0,00	0,08 0,03 $\pm$ 0,02	0,01 0,00 $\pm$ 0,00	0,00 0,00 $\pm$ 0,00	0,01 0,00 $\pm$ 0,01	0,13 0,00 $\pm$ 0,01	0,05 0,01 $\pm$ 0,01	0,06 0,02 $\pm$ 0,01	0,08 0,01 $\pm$ 0,01	0,04 0,00 $\pm$ 0,01	0,00 0,00 $\pm$ 0,00	0,00 0,00 $\pm$ 0,00	0,02 0,01 $\pm$ 0,01	0,05 0,01 $\pm$ 0,01	0,08 0,04 $\pm$ 0,02
isolet	0,52	0,41 0,31 $\pm$ 0,06	0,39 0,31 $\pm$ 0,06	0,44 0,30 $\pm$ 0,12	0,46 0,28 $\pm$ 0,14	0,27 0,17 $\pm$ 0,08	0,18 0,11 $\pm$ 0,05	0,54 0,29 $\pm$ 0,10	0,30 0,17 $\pm$ 0,08	0,55 0,32 $\pm$ 0,11	0,30 0,12 $\pm$ 0,06	0,44 0,25 $\pm$ 0,10	0,39 0,29 $\pm$ 0,10	0,45 0,36 $\pm$ 0,09	0,36 0,30 $\pm$ 0,06	0,57 0,24 $\pm$ 0,13	0,51 0,44 $\pm$ 0,09
AVG Max	0,23	0,23	0,23	0,21	0,22	0,19	0,12	0,39	0,23	0,25	0,23	0,37	0,20	0,24	0,22	0,39	0,43
Mean		0,20	0,20	0,16	0,16	0,17	0,09	0,17	0,17	0,14	0,08	0,21	0,15	0,18	0,18	0,16	0,37

Friedman Test

Table 7

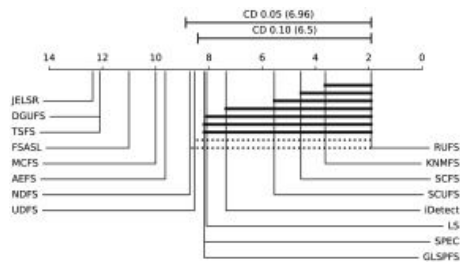
Average rank of methods (the smaller the better) by ACC, NMI and ARI (Max and mean). The best result are enlighten in bold and the second best is underlined.

Method	ACC Max	ACC Mean	NMI Max	NMI Mean	ARI Max	ARI Mean
LS	8,09	<u>4,91</u>	9,00	5,55	9,18	5,45
SPEC	8,18	5,09	9,00	6,18	9,09	5,64
MCFS	10,00	8,36	10,18	8,73	10,00	8,36
UDFS	8,55	8,91	7,45	8,36	8,82	10,09
JELSR	12,36	9,55	11,64	8,18	12,45	9,00
NDFS	8,73	10,55	12,73	12,64	13,09	13,09
RUFS	1,91	8,36	<u>2,55</u>	8,55	2,55	8,27
GLSPFS	8,18	8,09	<u>9,27</u>	7,45	8,27	7,27
FSASL	11,00	12,18	8,09	10,64	7,73	9,82
iDetect	7,36	12,27	8,55	11,91	9,18	12,27
SCUFS	5,55	8,55	6,18	9,8	4,91	8,91
DGUFs	12,09	9,91	11,91	10,00	11,91	10,18
AEFS	9,64	6,82	10,18	8,36	9,09	8,45
TSFS	12,09	9,36	10,36	7,73	10,18	7,18
SCFS	4,55	10,55	5,27	10,27	4,91	10,00
KNMFS	<u>3,64</u>	2,00	2,45	1,45	<u>3,27</u>	1,82

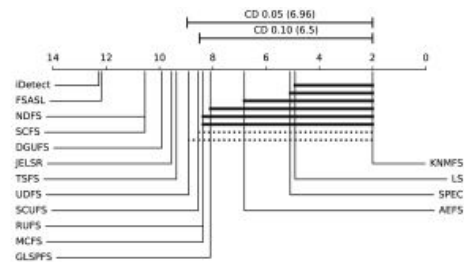
Table 8

Friedman test results.

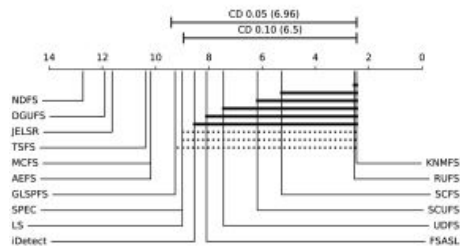
Metric	χ_F^2	F_F	Critical F	Test result
ACC Max	35,98	2,79	2,10	Reject null hypothesis
ACC Mean	48,86	4,21	2,10	Reject null hypothesis
NMI Max	51,11	4,97	2,10	Reject null hypothesis
NMI Avg	48,07	4,54	2,10	Reject null hypothesis
ARI Max	51,66	5,05	2,10	Reject null hypothesis
ARI Avg	50,71	4,91	2,10	Reject null hypothesis



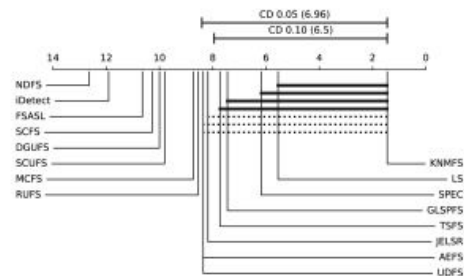
(a) ACC Max



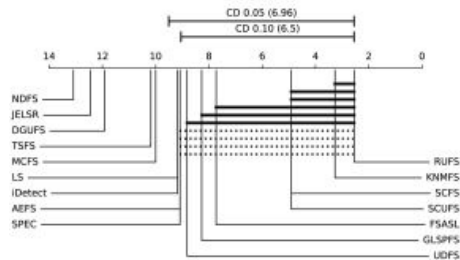
(b) ACC Mean



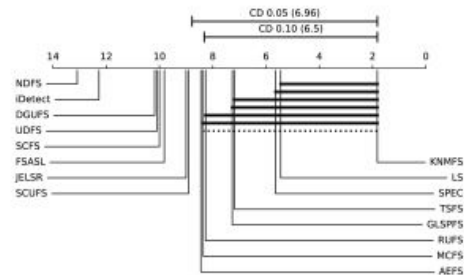
(c) NMI Max



(d) NMI Mean



(e) ARI Max



(f) ARI Mean



C . E . S . A . R

Pessoas impulsionando inovação.
Inovação impulsionando negócios.

Thanks!

